Isolation and purification

Topic Outline

- **Methods of Isolation and purification of the active constituents**
  - **Classical Methods**: Precipitation, Fractional crystallization, Fractional distillation, Chemical derivatization, Sublimation

Isolation and Purification

- Two general reasons for making separations on mixtures.

  To remove contaminants and/or pick the desired compound

  - **Isolation:** separating the desired drug from the rest of the mixture
  - **Purification:** removing substances considered to be contaminants

Methods of separation

S.No.	Method	Basis
1	Precipitation	Difference in solubilities
2	Distillation	Differences in volatility
3	Sublimation	Difference in vapour pressure
4	Extraction between two phases	Differences in solubility
5	Crystallisation	Property of solubility usually at low temperature
6	Flotation	Differences between the density of substance & liquid
7	Ultra filtration	Size of substance in comparison to filtering device
8	Dialysis	Osmosis-flow of system through a membrane
9	Electro deposition	Electrolysis at inert electrodes

To perform analysis

- Several impurities in a sample may interfere with the analysis of the substance of primary interest.
- Effective analysis can be achieved when impurities are removed
- Separation can also help make analysis of one or more components in a sample

- Heterogeneous groups of chemicals in crude extracts are separated as single components by different methods.
- These methods can be grouped as
  - **classical**
    - Precipitation, Fractional crystallization, Fractional distillation, Chemical derivatization, and Sublimation
  - **modern techniques**
    - the various chromatographic techniques

Separating insoluble solids from a liquid

Filtration

- Place a piece of filter paper in a funnel. Position a beaker under the funnel.
- Pour the mixture into the funnel. The liquid passes through the filter paper while the insoluble solids does not pass through.
- The filtrate will be collected in the beaker and the residue remains on the filter paper.
- Wash the residue with distilled water and let it dry. Separation is complete.

Precipitation

- Some compounds tend to undergo fractional liberation from a mixture.

  Examples:

  - Tannins can be precipitated by caffeine or lead acetate.
    - The tannins can be effectively separated from non-tannins using caffeine.

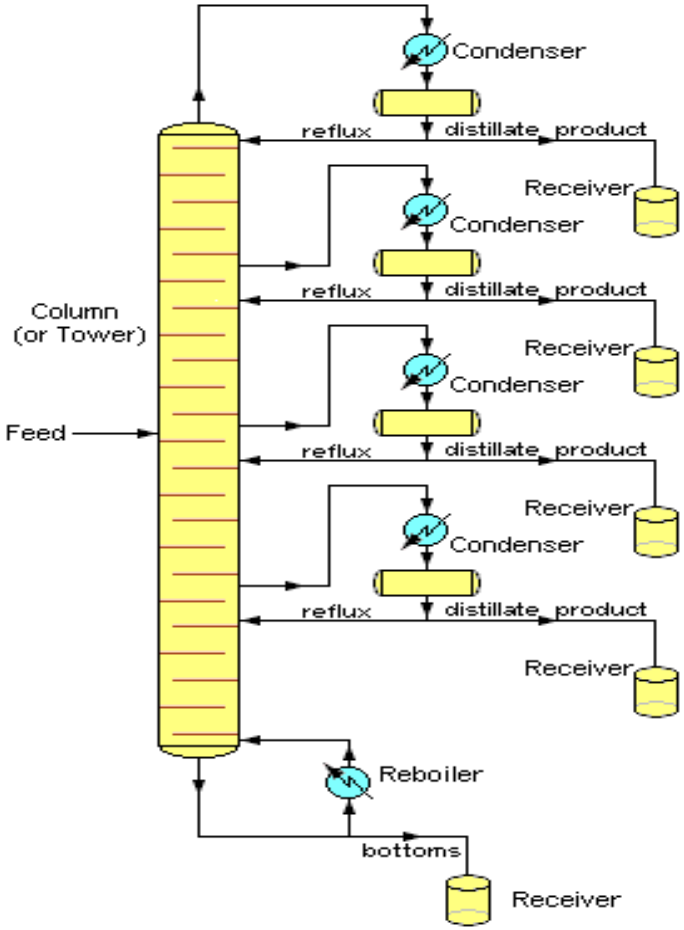
- A mixture of alkaloid salts in aqueous solutions,
  - When treated with aliquots of alkali, will give first the weakest base in the Free State followed by base liberation in ascending order of basicity.
  - If the mixture is shaken with an organic solvent after each addition, then a fractionated series of bases will be obtained.
- A similar scheme can be used for organic acids soluble in water-immiscible solvents.
  - in this case, starting with a mixture of the acid salts, it is possible to fractionally liberate the acids by addition of mineral acids

Fractional crystallization

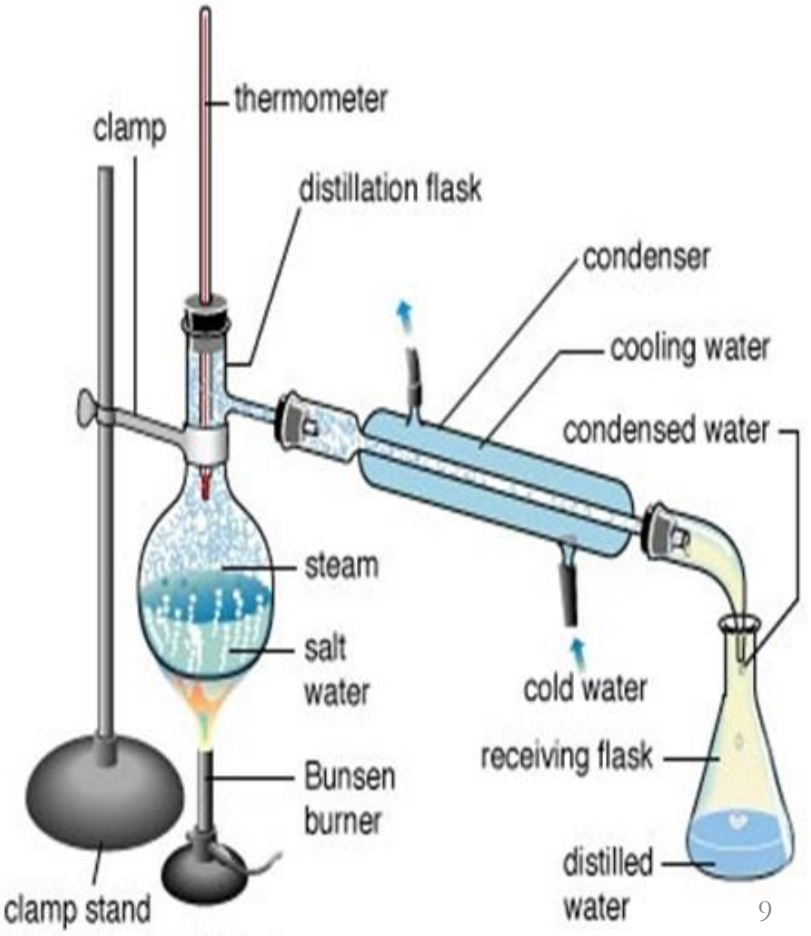
- ❖ This method exploits the differences in solubility of the components of a mixture in a particular solvent at low temperature.
- ❖ A process by which a chemical compound is separated into components by crystallization.
- ❖ In fractional crystallization the compound is mixed with a solvent, heated, and then gradually cooled so that, as each of its constituent components crystallizes, it can be removed in its pure form from the solution.
- Frequently, derivatives of the particular components are employed.
- Examples include
  - picrate of alkaloids
  - osazones of sugars

DISTILLATION

FRACTIONAL DISTILLATION: It is the traditional method of separation of constituents of volatile mixtures (isolation of components of volatile oils).



STEAM DISTILLATION: It is used to isolate volatile oils and hydrocyanic acid from plant material.



Fractional distillation

- This process can be used to separate miscible liquids with different boiling points
- Liquid with lower boiling point will vaporise first
- e.g. to separate alcohol and water
- Employs a distillation apparatus with a fractionating column
- In phytochemistry, it is widely used for the isolation of the components of volatile oils
- The different components are separated in the fractionating column based on their boiling point difference

Sublimation

- Process by which a substance changes from solid state to vapour state on heating without undergoing intermediate liquification.

  - e.g of substances which sublime

  - iodine

  - ammonium chloride



- On heating, they do not melt. Iodine changes into a beautiful violet vapour while ammonium chloride changes into a white vapour.
- They change back into solid crystals on cooling.
- Process used to separate a solid that sublimes from one that does not

e.g. iodine or ammonium chloride from common salt

- May sometimes be possible for the whole drug,
  - as in the isolation of caffeine from tea or
  - for the purification of materials in a crude drug
- Modern techniques employ low pressure with a strict control of temperature
- When a mixture of a solid that sublimes and a solid that does not is heated, the solid that sublimes will turn into a vapour and separates from the other solid which remains in the container.

Modern techniques/Chromatographic techniques

- **Chromatography**
  - Is essentially a group of techniques for the separation of the compounds of mixtures by their continuous distribution between two phases one of which is moving past the other.
  - It is a broad range of physical methods used to separate and/ or to analyze complex mixtures.
  - The components to be separated are distributed between two phases:
    - a *stationary phase* bed and
    - a *mobile phase* which percolates through the stationary bed.

What is Chromatography?

Chromatography is a technique for separating mixtures into their components in order to analyze, identify, purify, and/or quantify the mixture or components.

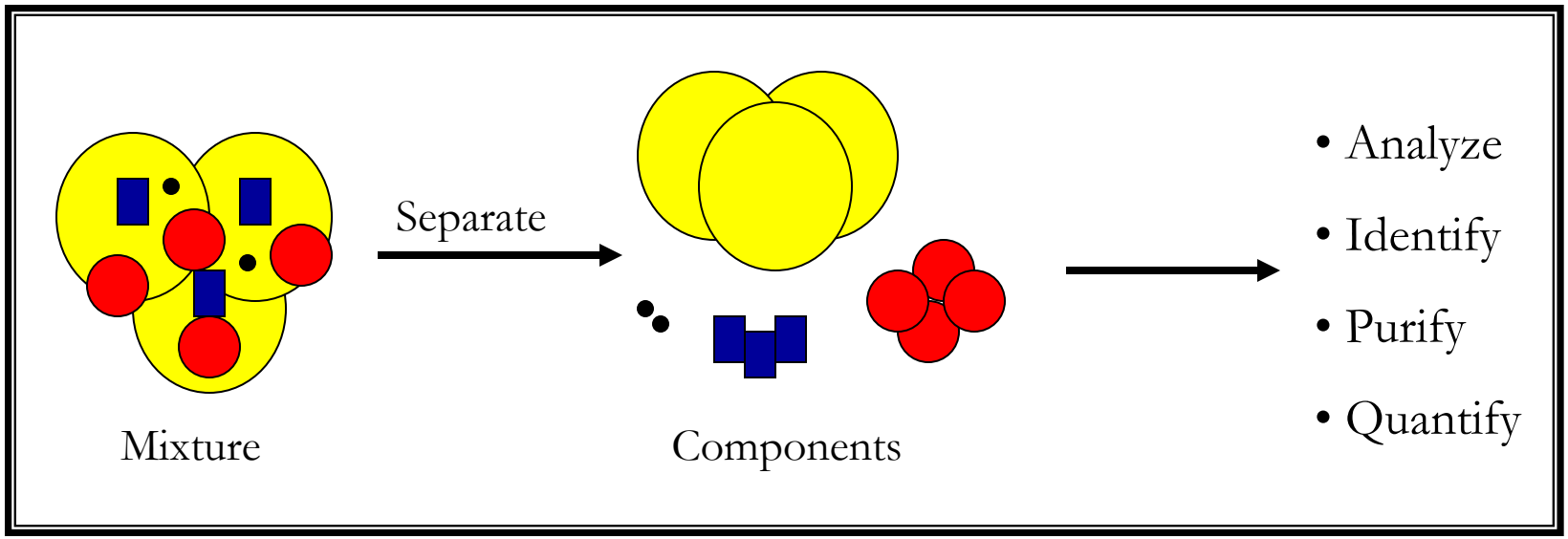
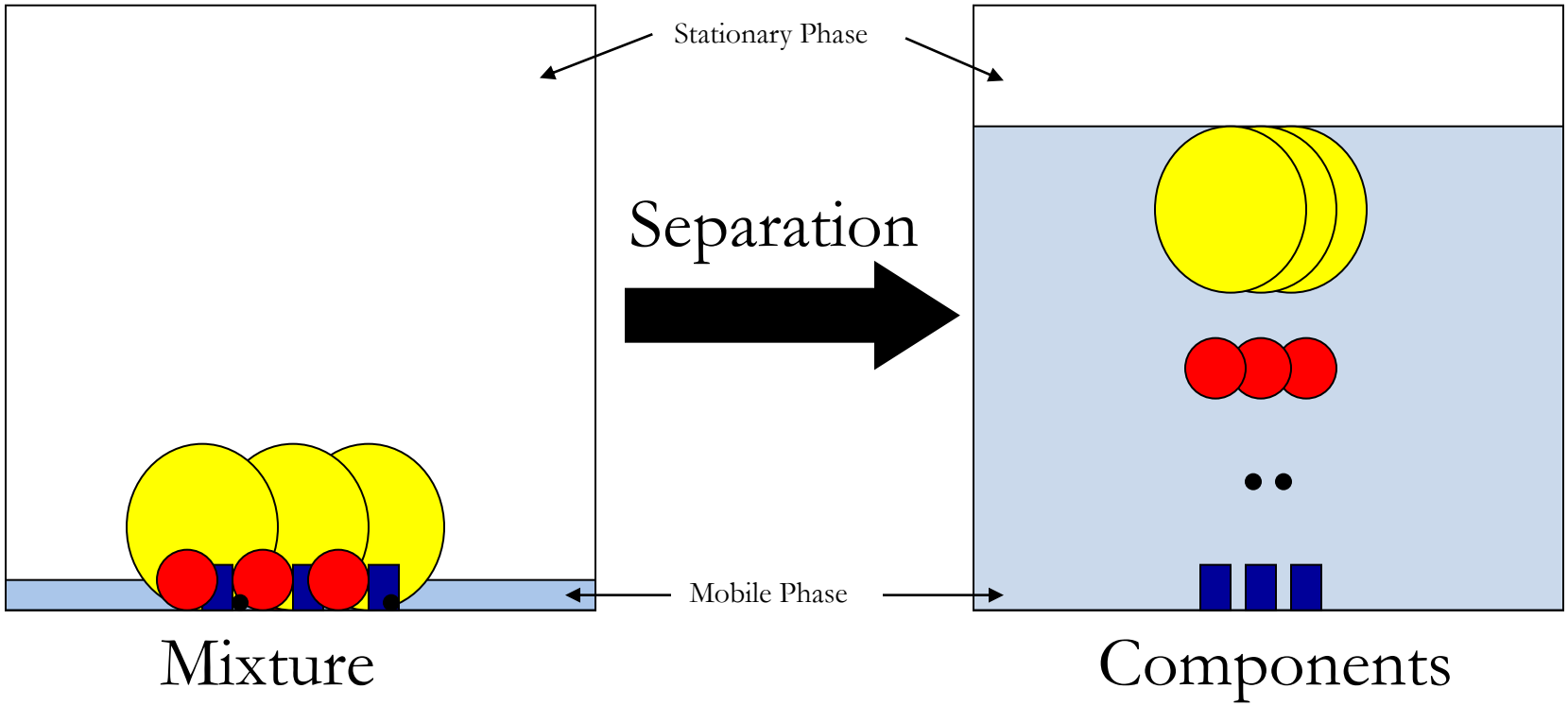


Illustration of Chromatography



Components	Affinity to Stationary Phase	Affinity to Mobile Phase
Blue	----------	Insoluble in Mobile Phase
Black	✓ ✓ ✓ ✓ ✓ ✓	✓ ✓
Red	✓ ✓	✓ ✓ ✓ ✓ ✓
Yellow	✓ Dr.YR	✓✓✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓

Uses for Chromatography

Chromatography is used by scientists to:

- Analyze – examine a mixture, its components, and their relations to one another
- Identify – determine the identity of a mixture or components based on known components
- Purify –separate components in order to isolate one of interest for further study
- Quantify – determine the amount of the a mixture and/or the components present in the sample

Uses for Chromatography

Real-life examples of uses for chromatography

- Pharmaceutical Company – determine amount of each chemical found in new product
- Hospital – detect blood or alcohol levels in a patient's blood stream
- Law Enforcement – to compare a sample found at a crime scene to samples from suspects
- Environmental Agency – determine the level of pollutants in the water supply
- Manufacturing Plant – to purify a chemical needed to make a product

Definition of Chromatography

Detailed Definition:

Chromatography is a laboratory technique that separates components within a mixture by using the differential affinities of the components for a mobile medium and for a stationary adsorbing medium through which they pass.

Terminology:

- *Differential* – showing a difference, distinctive
- *Affinity* – natural attraction or force between things
- *Mobile Medium* – gas or liquid that carries the components (mobile phase)
- *Stationary Medium* – the part of the apparatus that does not move with the sample (stationary phase)

Definition of Chromatography

Simplified Definition:

Chromatography separates the components of a mixture by their distinctive attraction to the mobile phase and the stationary phase.

Explanation:

- Compound is placed on stationary phase
- Mobile phase passes through the stationary phase
- Mobile phase solubilizes the components
- Mobile phase carries the individual components a certain distance through the stationary phase, depending on their attraction to both of the phases

Chromatography terms

- The **analyte** is the substance that is to be separated
- **Analytical chromatography** is used to determine the existence and possibly also the concentration of analyte(s) in a sample.
- **Preparative chromatography** is used to purify sufficient quantities of a substance for further use, rather than analysis.
- A **bonded phase** is a stationary phase that is covalently bonded to the support particles or to the inside wall of the column tubing.

- A **chromatogram** is the visual output of the chromatograph. In the case of an optimal separation, different peaks or patterns on the chromatogram correspond to different components of the separated mixture.
- The **retention time** is the characteristic time it takes for a particular analyte to pass through the system (from the column inlet to the detector)
- The **sample** is the matter analysed in chromatography
- The **stationary phase** is the solid substance which is fixed in place for the chromatography procedure. Examples include the silica or calcium carbonate layer.
- The **mobile phase** is the liquid phase which moves in a definite direction
- The **solvent** refers to any substance capable of solubilizing other substance, and especially the liquid mobile phase in LC.

- The **solute** refers to the sample components in partition chromatography.
- A **chromatograph** is equipment that enables a sophisticated separation e.g. gas chromatographic or liquid chromatographic separation.
- **Chromatography** is a physical method of separation in which the components to be separated are distributed between two phases, one of which is stationary (stationary phase) while the other (the mobile phase) moves in a definite direction.
- The **effluent** is the mobile phase leaving the column.
- An **immobilized phase** is a stationary phase which is immobilized on the support particles, or on the inner wall of the column tubing.

How Does Chromatography Work?

- A mixture of various components enters a chromatography process, and
  - flushed through the system at different rates.
- As the mixture moves over a stationary phase differential rates of migration provide separation.
- Repeated sorption/desorption events that take place during the movement of the sample over the stationary bed determine the rates.
- The smaller the affinity a molecule has for the stationary phase, the shorter the time spent in a column.

- Chromatography is a *very* special separation process for a multitude of reasons!
  - First of all,
    - It can separate complex mixtures with great precision.
    - Even very similar components, such as proteins that may only vary by a single amino acid, can be separated.
    - In fact, it can purify basically any soluble or volatile substance if the right adsorbent material, carrier fluid, and operating conditions are employed.
  - Second,
    - It can be used to separate delicate products since the conditions under which it is performed are not typically severe.
    - For these reasons, chromatography is quite well suited to a variety of uses

Choosing Adsorbents and Eluents

- **Stationary phase: Adsorbents.**
  - Alumina is generally used for chromatography of less polar compounds.
  - Silica gel is better for compounds containing polar functional groups.



- **Mobile phase: Eluents** (liquid)
  - The more polar eluent = the greater eluting power: its ability to move compounds over the adsorbent surface.
  - Mixed eluents can be prepared by mixing low polarity and high polarity solvents and therefore creating any eluting power needed.

least polar — cyclohexane
petroleum ether
hexane
toluene
dichloromethane
ethyl acetate
ethanol
acetone
most polar — methanol

Chromatographic methods

S.No	Method	Basis
1	Adsorption column chromatography	Distribution of solute between a solid and liquid phase in a column
2	Partition column chromatography	Distribution of a solute between two liquids on a column
3	Paper chromatography	Partition on a sheet of paper
4	Thin Layer chromatography	Adsorption or partition on thin sheets
5	Ion exchange chromatography	Exchange of ion
6	Molecular sieves	Size of solution
7	Gas chromatography	Distribution of gaseous solutes between a gas & liquid or solid phase
8	Zone electrophoresis	Separation on a sheet in presence of an electric field

Types of Chromatography

- Based on the modes (mechanisms) of separation, chromatography can be:
  - Adsorption
  - Partitioning
  - Exclusion
  - Ion Exchange
  - Affinity

Classification of Chromatography

- **Based on** the technical application of the stationary phase, chromatography can be:
  - **Columnar type of Chromatography**
    - The stationary phase is held in a narrow tube through which the mobile phase is forced under pressure or by gravity.
  - **Planar type of Chromatography**
    - The stationary phase is supported on a flat plate or the interstices of a paper and
    - The mobile phase moves through the stationary phase by capillary action or by gravity.

- **Based on** the type of mobile phase, chromatography can be:
  - **Liquid chromatography**

    Includes
    - Column chromatography (CC),
    - Thin-layer chromatography (TLC), and
    - High performance liquid Chromatography (HPLC)
- Stationary phase:
  - Silica, alumina, etc.
- Mobile phase (moving phase):
  - organic solvents
  - **Important properties:**
    - Polarity

- **Gas chromatography**
  - Stationary phase:
    - A film of a polymer or a wax.
    - The film must have a high boiling point
  - Mobile phase:
    - gas (Helium is the usual carrier gas)
  - Important properties:
    - boiling point
- **Theoretical concepts of Chromatography**
  - In a typical chromatographic system, different compounds move through the system
    - With different rates of movement(differential migration)
    - In the form of zones or bands

Column Chromatography

- It is also known as classical (traditional) chromatography
  - This is simply to differentiate it from other advanced techniques
- It utilizes a vertical glass column filled with solid support with the sample on top of this support.
- The rest of the column is filled with a solvent which, under the influence of gravity, moves the sample through the column.
- The stationary phase (column packing in the column) is very polar.
- One starts with a less polar solvent to remove the less polar compounds, and then slowly increase the polarity of the solvent to remove the more polar compounds.

- **Polar compounds**
  - Are going to be attracted to the polar column packing by
    - hydrogen bonding or
    - Dipole-dipole attractions
  - Are going to move slowly!
- **Non-polar compounds** elute first, while the polar compounds elute off the column later.

- **Development and application**
  - **Column packing**
    - Should be Uniformly packed
  - **Bottom of column**
    - Sintered glass or
    - cotton plug and sand layer

- **Adsorbent packed in the form of**
  - Slurry (for non dense adsorbent)
  - Powder (for dense adsorbent as it create difficulty of slurry formation) usually with tapping
- **Top of column**
  - Filter paper plus sand (to avoid disturbance of adsorbent while adding mobile phase)
- **Sample applied at the top as**
  - Powder
  - paste or
  - solution
- **Solvent allowed to percolate**
  - It should not be allowed to fall below the level of the adsorbent (to avoid problem of cracking of adsorbent)

- The complete separation of all constituents into distinct band does not result during the 1st adsorption stage, but the bands remain crowded together near the top of the column.
- Such a column may be developed by allowing more pure solvent to percolate through the column when the adsorbed materials slowly pass downwards & the separate bands become wider.
- In many cases the process can be carried out efficiently by using a different solvent, one from which the substances are less strongly adsorbed.
- Substances are more readily adsorbed from non-polar solvents such as light petroleum & benzene.
- While polar solvents like alcohol, water & pyridine are useful eluting media.

- Many substances are adsorbed at one pH & eluted at another.
- Various substances may be used as adsorbing materials.
- Ex. Alumina, Kaolin, Magnesium oxide, calcium carbonate, charcoal & sugar. Powdered sucrose was used by Tswett for the separation of chlorophyll a & b.
- When colourless substances are chromatographed, the zone may not be visible but shows fluorescent zone when examined under U.V light.

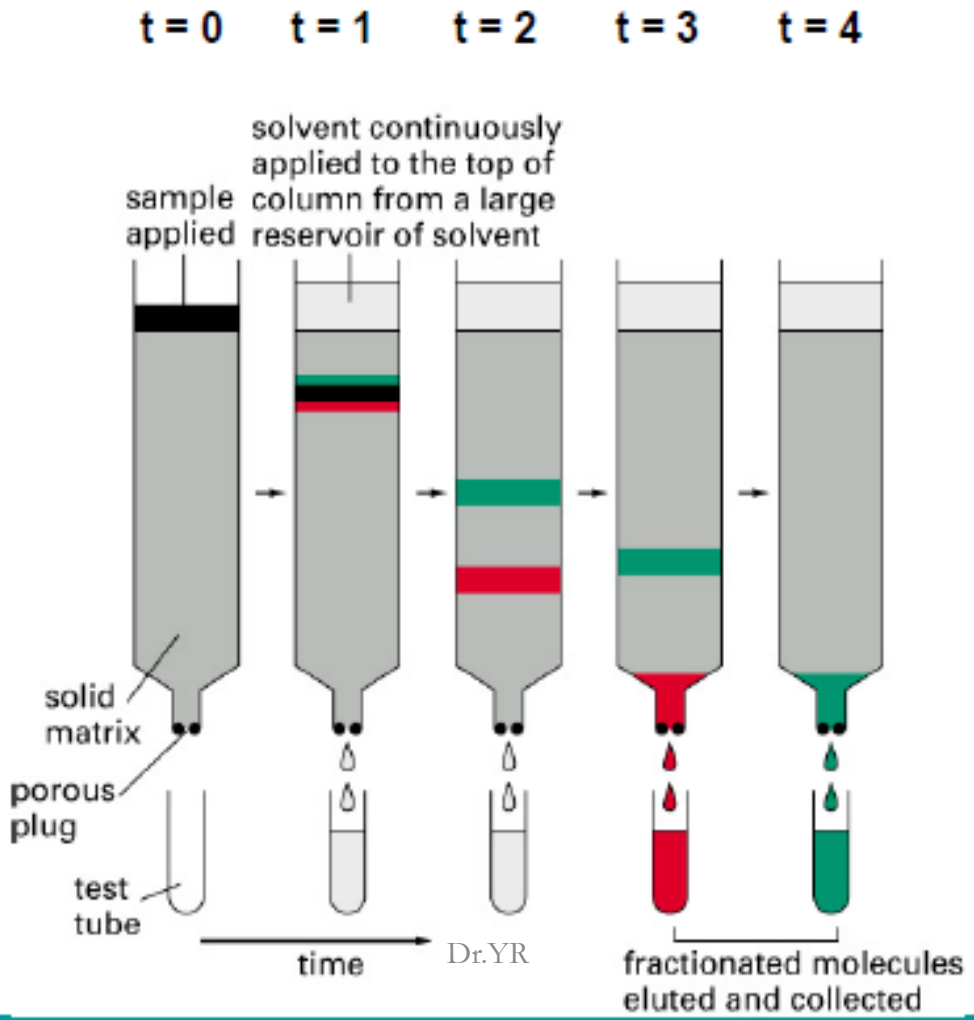
  Ex. Vitamins, hormones, many alkaloids,

  cardiac glycosides, anthraquinones etc.



Column Chromatography

- Column chromatography: technique that uses an adsorbent packed in a glass column, and a solvent that moves down slowly through the packed column.



Planar Chromatography

- It is a chromatographic technique in which the supporting medium is a flat bed or plane rather than a column
- The stationary phase is present as or on a plane
- The plane can be a paper, (paper chromatography) or a layer of solid particles spread on a support such as a glass plate (thin layer chromatography)
- Different compounds in the sample mixture travel different distances according to how strongly they interact with the stationary phase as compared to the mobile phase
- The specific Retardation factor (R_f) of each component can be used to aid in the identification of an unknown substance.
  - separation can be based on either adsorption or partitioning.

Planar Chromatography

- **Some differences between column and planar chromatography**

1. Presence of a third phase in the planar system.

- This third phase is the atmosphere or gaseous vapours located above the surface.
- In order to obtain reproducible separation, it is necessary to control this phase as well as the stationary and mobile phases.

Planar Chromatography

2. Means by which solutes are separated

- **Planar chromatography** separates solutes based on their d/t distances of travel in a set amount of time
- **Column chromatography** separates solutes by their d/t times of travel along a given distance, the length of the column.

Planar Chromatography

- Linear development may be performed as
  - **Ascending** development
    - with the sample and **mobile phase** applied at the bottom of the support,
  - **Descending** development
    - with the sample and **mobile phase** applied at the top of the support, or
  - **Horizontal** development
    - with the sample and mobile phase applied to the edge of a support lying flat.

Planar Chromatography

- **Sample application**
  - Is an important feature to obtain good separations
  - Is generally done by
    - placing the solute mixture in a volatile solvent and
    - applying a small amount of the mixture to the support.
    - solvent is allowed to evaporate, leaving solute on the plate.
  - In HPTLC, sample spots less than 1 mm width is needed.
  - Automatic systems such as robots are typically used.

Planar Chromatography

- **Detection**
  - Some spots can be visible in the daylight.
  - But, most of the time, the spots are not visible
  - Visualization is a method that is used to render the TLC spots visible. A visualization method can be:
    - Ultraviolet light
    - Iodine vapours to stain spots
    - Coloured reagents to stain spots
    - Reagents that *selectively* stain spots while leaving others unaffected.
  - Thus , detection can be made through:
    - Direct measurement;
    - Chemical derivatization.

Planar Chromatography

1. **Direct measurement**

Florescence:

- Emitting light or quenching fluorescence from the substrate.
  - ✓**Radioactivity** of compounds containing radioisotopes in their structure can also be used to identify the presence of solute bands on a plate.
    - This can be done by using radioactive detectors
  - ✓**Instrumentation detection**

2. Chemical derivatization

- ✓Iodine adsorption.
- ✓Oxidation using sulfuric acid

Types of planar Chromatography

- **Paper chromatography (PC)**
  - Paper(cellulose) is used as both the support material and stationary phase
  - The mechanism of separation is primarily partition
- **Thin-layer chromatography (TLC)**
  - Use any non-paper materials as the supporting medium.
  - Conventional TLC and
  - High performance TLC (HPTLC)

Paper Chromatography

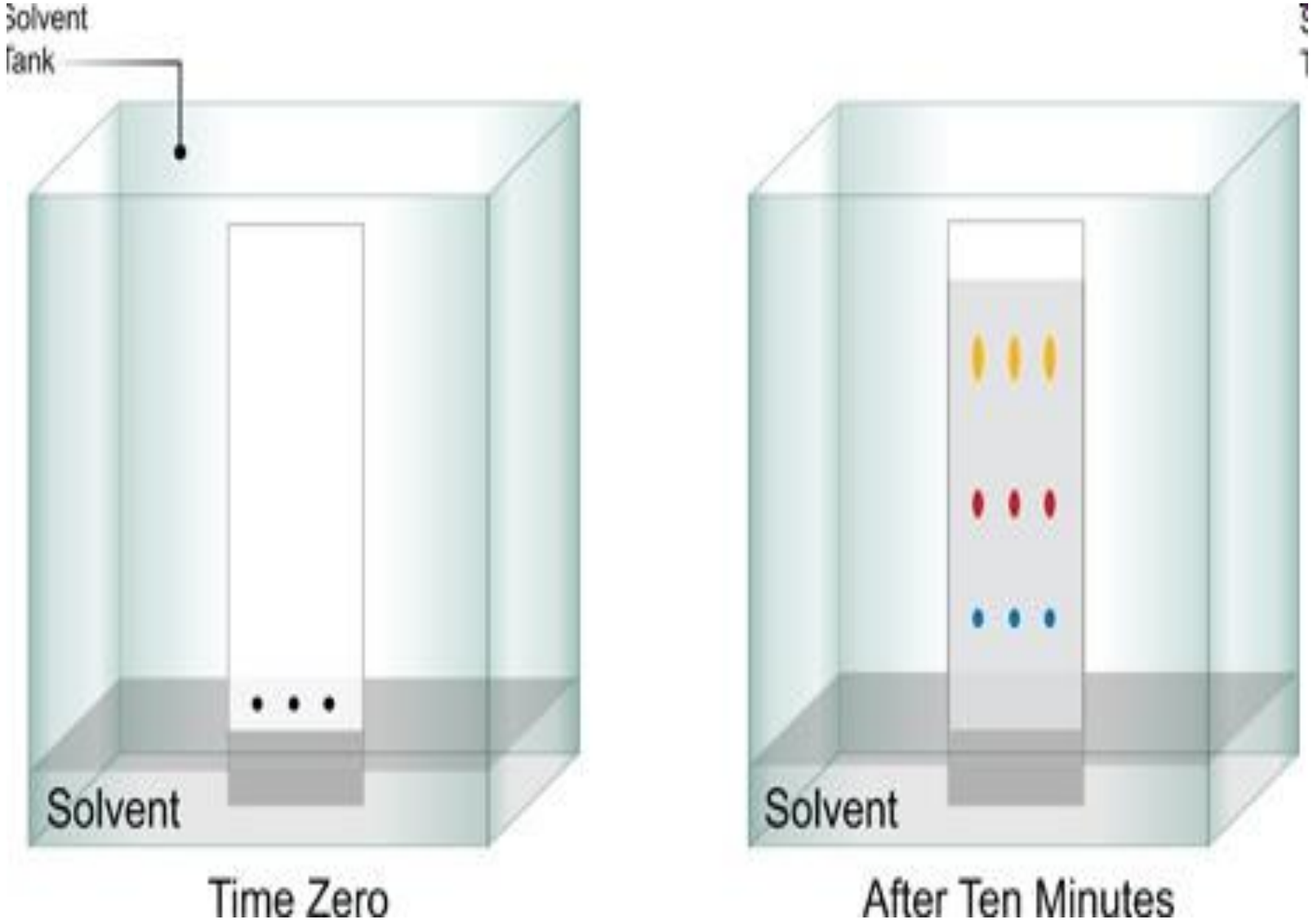
- It is a technique that involves placing a small spot near one end onto a strip of *chromatography paper*.
- Paper chromatography is a useful technique because it is relatively quick and requires small quantities of material.
- The paper is placed in a air tight chamber saturated with a shallow layer of solvent & water.
- As the solvent rises through the paper, it meets the sample mixture which starts to travel up the paper with the solvent

- The most satisfactory solvents are those which are partially miscible with water, phenol, n-butanol & amyl alcohol.
- The paper is developed by dipping in the solvent in different ways of solvent movement (ascending, descending, radial & two dimmensional).
- As the solvent move, the components move along the paper at varying rates depending on their partition co-efficients between the aqueous (hydration chell of cellulose fibres) & organic phases.
- After the paper strip is dried, the positions of the separated components can be seen by using suitable developing agents.

S.No	Detecting agent	Components
1	Ninhydrine solution	Amino acids
2	Iodine solution or vapours	Alkaloids
3	$Fecl_3$	Phenols
4	Alkali	Anthraquinone derivatives
5	Antimony chloride in chloroform	Steroids & some components of volatile oils
6	Aniline hydrogen pthalate	Sugars

- Improved separation of mixtures can be obtained by adjusting the acidity of the solvent with NH_3, CH_3COOH, HCl or by impregnating the paper with buffer solution or formamide solution.
- Drugs which are evaluated by this technique include aloes, ergot, nux-vomica, opium, rauwolfia, rhubarb, volatile oils etc.

- More polar substances bond with the cellulose paper more quickly, and therefore do not travel as far.

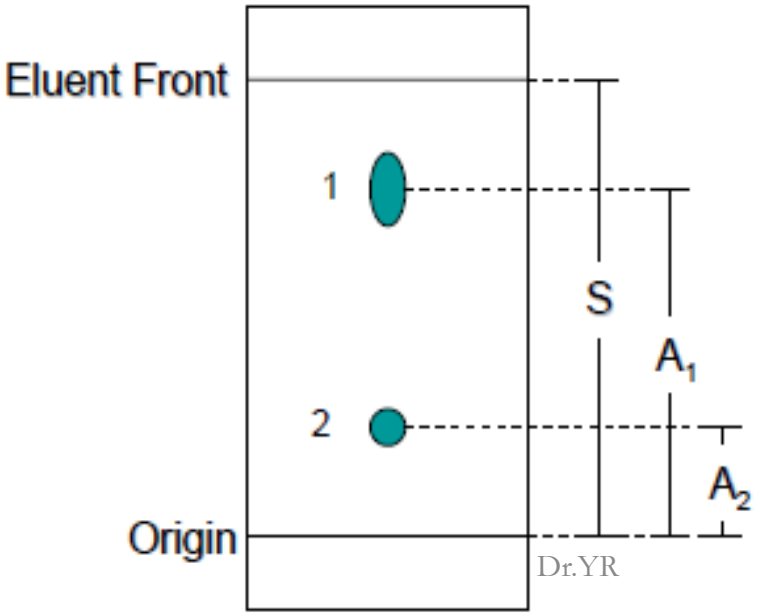




Retention Factor (R_f)

- Definition: ratio of the distance that a compound moves to the distance that the eluent front moves.

$$R_f = \frac{\text{distance traveled by compound, mm}}{\text{distance traveled by eluent front, mm}}$$



$$Rf_1 = \frac{A_1}{S}$$

$$Rf_2 = \frac{A_2}{S}$$

Thin-layer Chromatography (TLC)

- It was first demonstrated by stahl in 1958.
- It involves the use of any non-paper materials as the supporting medium.
- A thin layer of adsorbant material on a suitable glass plate, are essentially the same as used in column liquid chromatography.
- These can include materials used in
  - Adsorption chromatography (silica and alumina),
  - Liquid coated support or bonded-phases for use in partition chromatography (reversed-phase and normal-phase), and
  - Support for use in ion-exchange, affinity and size exclusion chromatography.

- The mobile phase used in TLC
  - Is also essentially the same as used in column liquid chromatography
  - The exactly used type will depend on the stationary phase and type of separation to be performed.
- The mixture to be resolved are dissolved in a suitable solvent and placed as a spot by marking the origin lineacross the plate 1cm from the bottom at one end of the plate & dipped in a suitable solvent mixture & the whole enclosed in an airtight container.
- Make sure your spots are above the solvent level in the tank, then carefully put the plate in the tank.

- The solvent(eluent) travels up the plate & after sometime when it reaches 2/3rd of the plate it is removed, the solvent front marked.
- Then the solvent allowed to evaporate & the positions of the separate compounds can be observed by spraying detectors.
- Compared to paper, it has the advantage of faster run, better separation, and the choice between different adsorbents. For even better resolution and to allow for quantitation, high-performance TLC can be used.

- TLC the less polar mobile phase (the eluent or eluting solvent) is an organic solvent and the stationary phase is a more polar solid such as silica gel (SiO_2) or alumina
- To visualize your spots check the plate in the UV light box using the short-wavelength lamp. Circle any spots and **turn off the lamp when you are done**.
- Our stationary phase is silica gel which is very polar.
  - Compounds with smaller R_f values are more polar because they have greater affinity for the polar stationary phase, and therefore, hang closer to the baseline.
  - Conversely, compounds with larger R_f values are less polar because they have greater affinity for the non-polar mobile phase, and travel with it further up the plate.

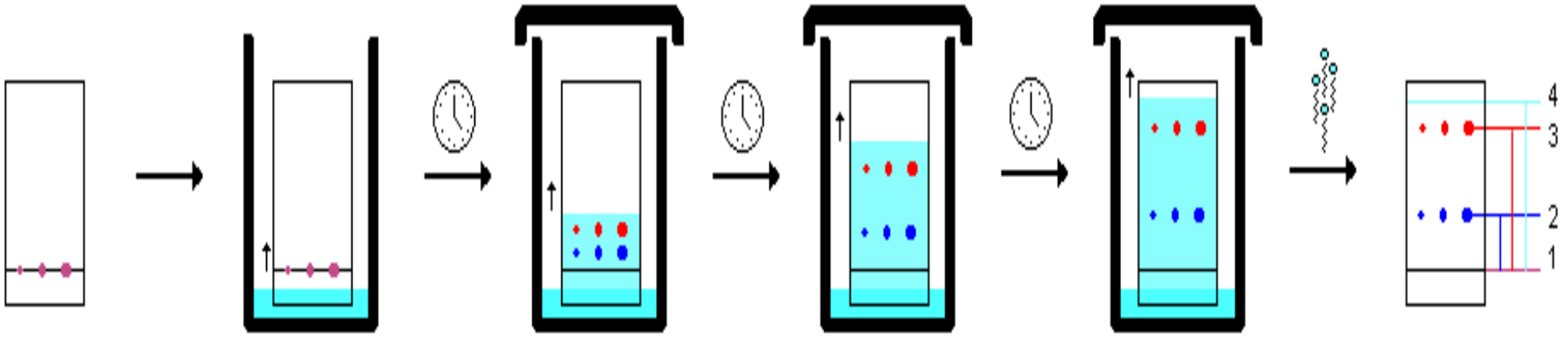
Identifying the Spots

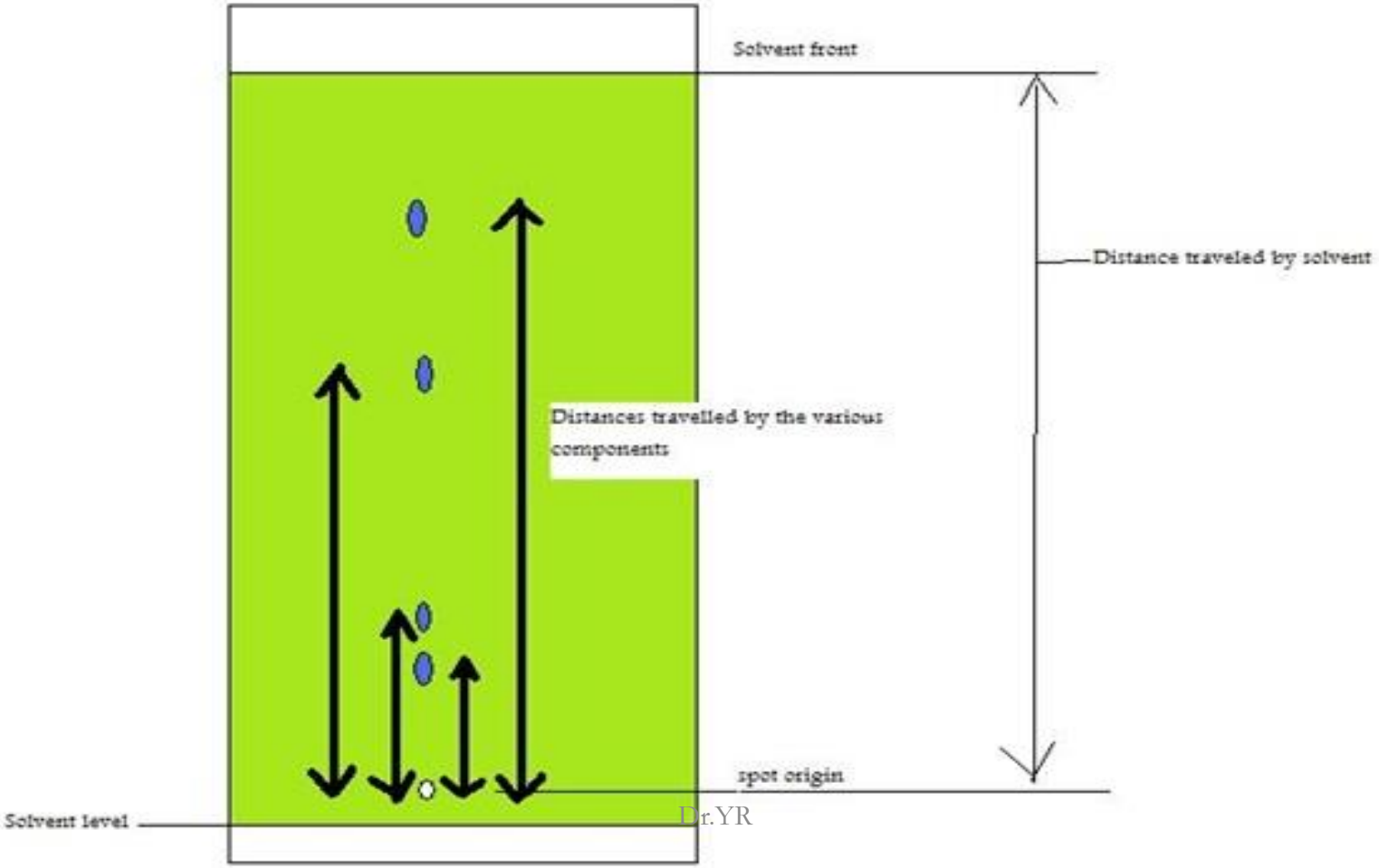
- If the spots are visible under normal light,
  - ➢outline them with a pencil.
- If no spots are visible under normal light, the most common visualization technique is
  - ➢to hold the plate under a UV lamp
    - o (CAUTION: Do not look directly into the lamp.)
- Many organic compounds can be seen using this technique, and
- many commercial plates often contain a substance which aids in the visualization of compounds.

TLC is best used for moderately or weakly polar mixtures.

Stationary phase and mode of separation

Stationary Phase	Chromatographic Mechanism	Typical Application
Silica Gel	adsorption	steroids, amino acids, alcohols, hydrocarbons, lipids, aflaxtoxin, bile, acids, vitamins, alkaloids
Silica Gel RP	reversed phase	fatty acids, vitamins, steroids, hormones, carotenoids
Cellulose, kieselguhr	partition	carbohydrates, sugars, alcohols, amino acids, carboxylic acids, fatty acids
Aluminum oxide	adsorption	amines, alcohols, steroids, lipids, aflatoxins, bile acids, vitamins, alkaloids
PEI cellulose	ion exchange	nucleic acids, nucleotides, nucelosides, purines, pyrimidines
Magnesium silicate	adsorption	steroids, pesticides, lipids, alkaloids


4
3
2
1


Solvent front
Distance traveled by solvent
Distances travelled by the various components
spot origin
Solvent level

Advantages of TLC

- TLC is very simple to use and inexpensive.
- There are little materials needed for TLC (chamber, watch glass, capillary, plate, solvent, pencil, and UV-light).
- More than 1 compound can be separated on a TLC plate as long as the mobile phase is preferred for each compound.
- The solvents for the TLC plate can be changed easily and it is possible to use several different solvents depending on your desired results. As stated earlier, TLC can be used to ensure purity of a compound.
- It is very easy to check the purity using a UV-light. Identification of most compounds can be done simply by checking R_f literature values.
- You can modify the chromatography conditions easily to increase the optimization for resolution of a specific component.

Disadvantages of TLC

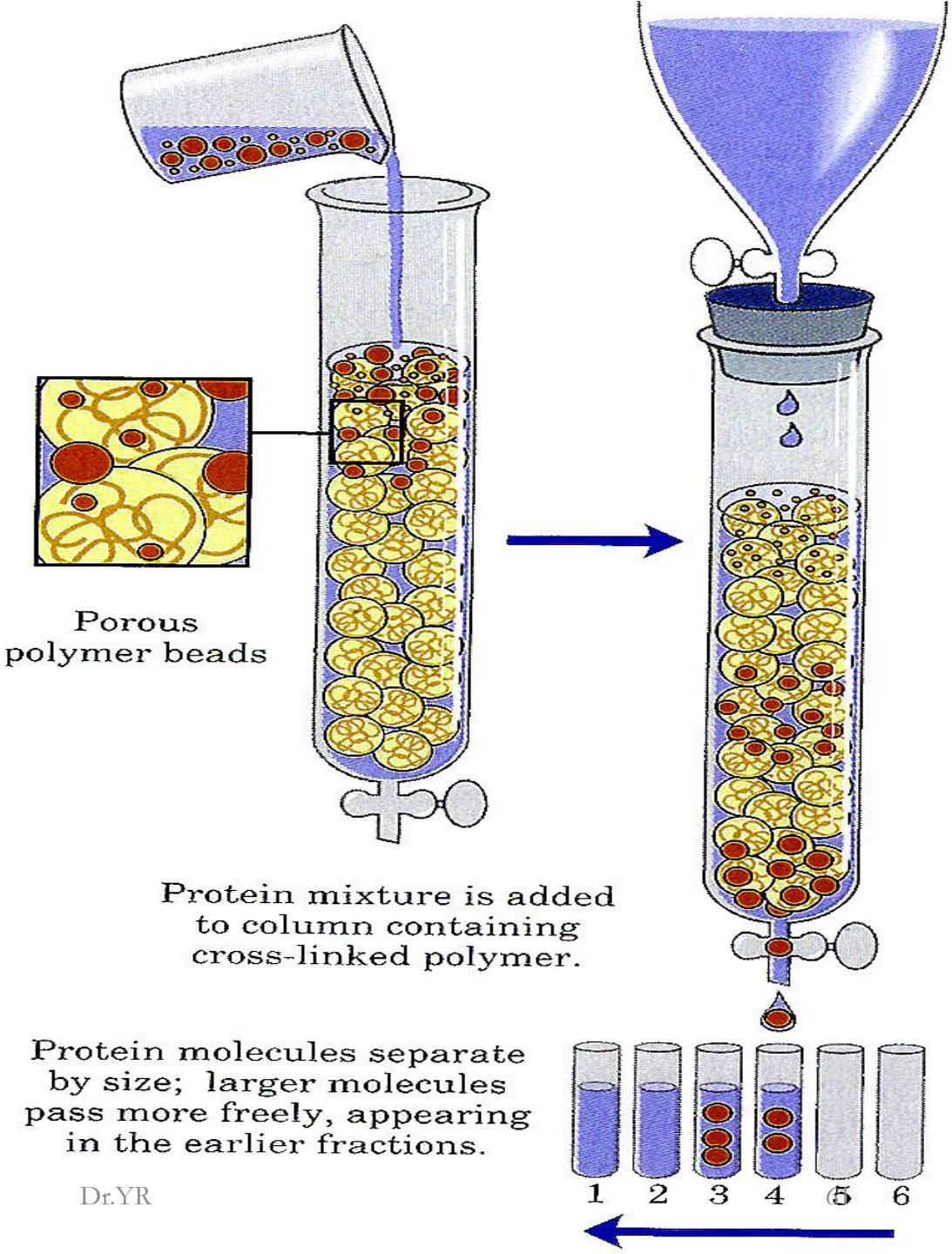
- TLC plates do not have long stationary phases. Therefore, the length of separation is limited compared to other chromatographic techniques. Also, the detection limit is a lot higher. If you would need a lower detection limit, one would have to use other chromatographic techniques.
- TLC operates as an open system, so factors such as humidity and temperature can be consequences to the results of your chromatogram.

Gel Chromatography

- Size exclusion chromatography (SEC) is also known as **gel permeation chromatography** (GPC) or **gel filtration chromatography** and separates molecules according to their size (or more accurately according to their hydrodynamic diameter or hydrodynamic volume).
- It is also useful for determining the tertiary structure and quaternary structure of purified proteins, especially since it can be carried out under native solution conditions.
- It is generally a low resolution chromatography technique and thus it is often reserved for the final, "polishing" step of a purification.

- Hydrophilic gels such as those prepared from starch, agar, agarose, polyacrylamide, polyvinyl carbitol & cross linked dextrans(sephadex) have been used for the fractionation of proteins, peptides, aminoacids & polysaccharides.
- The particles of gels possess pores formed by the molecular structures of the gel, then packed in a column & percolated with a solution.
- Smaller molecules are able to enter the pores of the media and, therefore, take longer time to elute, whereas larger molecules are excluded from the pores and elute faster.

Size-exclusion Chromatography



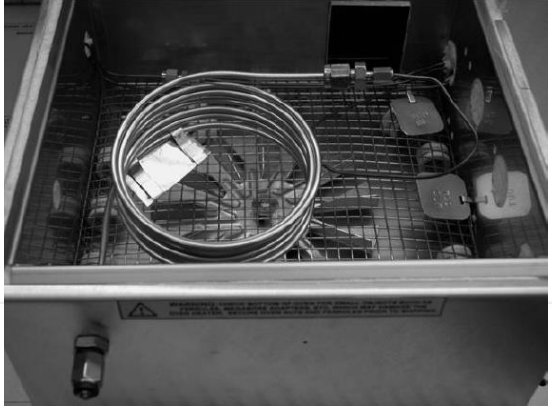
Gas Chromatography

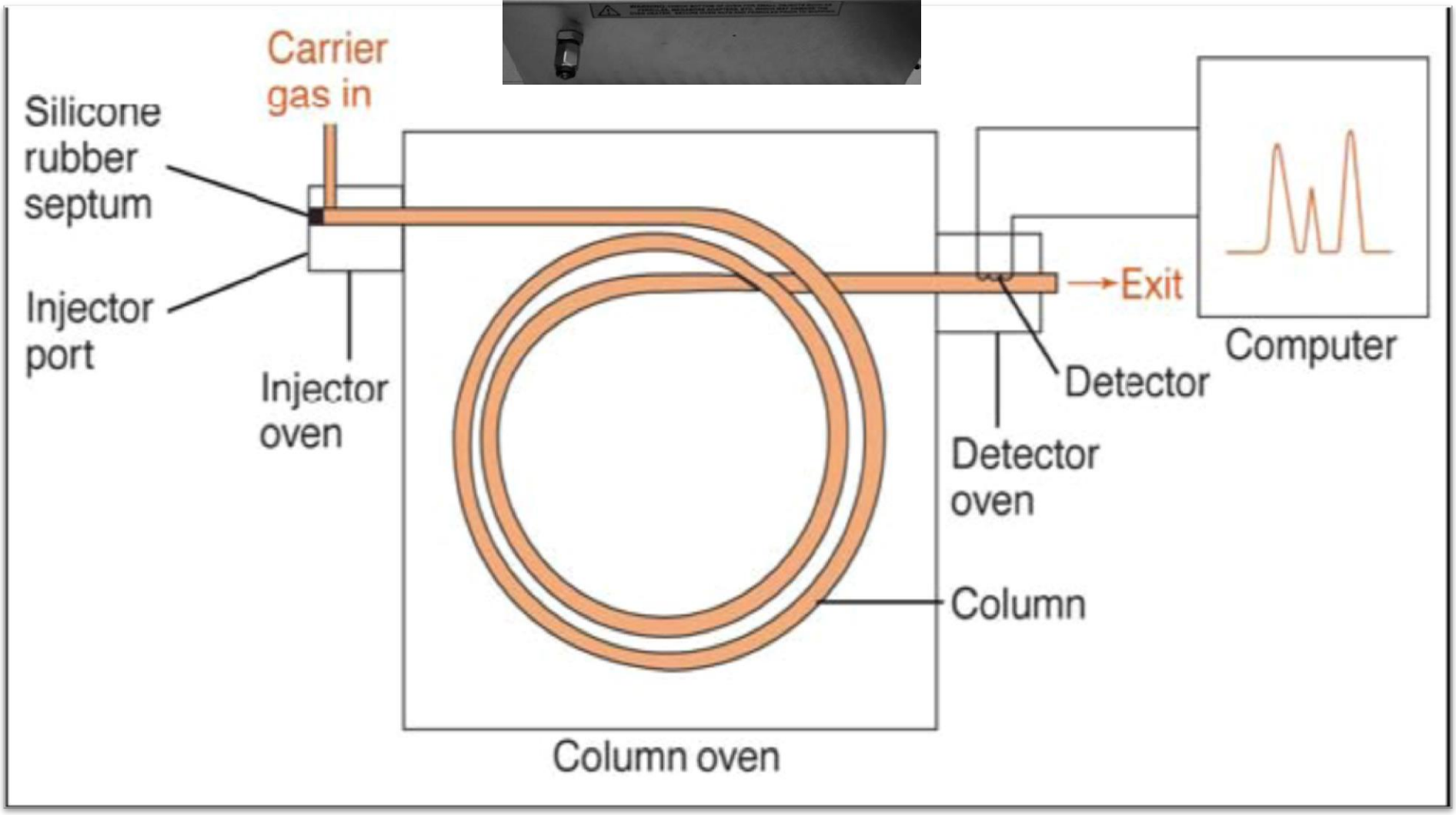
- **Gas Chromatography** (GC) is a common technique used to separate and identify volatile organic compounds (analytes). As the gas moves the analyte across the stationary phase, the analyte will be in equilibrium with the gas and the liquid phase (usually suspended on a solid surface).
- This method depends upon the **solubility** and the **boiling point** of the volatile organic liquid in order to separate them from a mixture.
- It's both a **qualitative** (identity) and **quantitative** (how much) tool.
- The **mobile phase** is an inert gas. Commonly used gases include N_2, He, Ar, and CO_2, depended on the type of detector.
- The **stationary phase** is a high-boiling liquid film supported on an inert solid and packed in either a fused silica (12 to 30 meter) capillary column or in a copper or stainless steel (1.5 to 3.0 meter) metal column.
- Derivatisation of a sample is carried out to
  - reduce polarity of the analytes
  - increase the detector response
  - increase volatility of the analytes

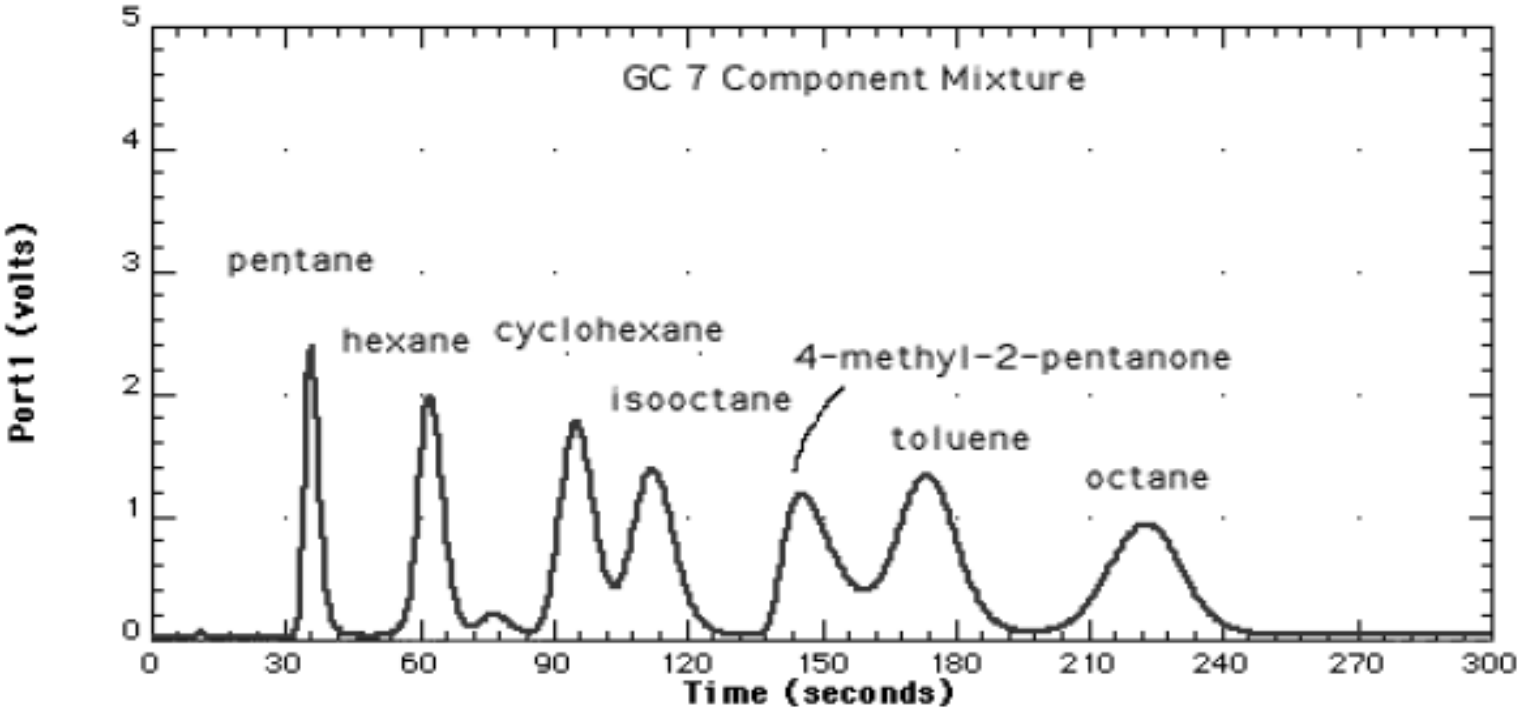
- Is a process by which a mixture is separated into its constituents by moving gas phase passing over a stationary phase.
- Is a useful technique
  - for substances that do not decompose at high temperatures and
  - when a very small quantity of sample (micrograms) is available
- Based on the nature of the stationary phase
  - Gas-liquid chromatography(GLC)
  - Gas-solid chromatography (GSC)
- Molecules with greater affinity for the stationary phase spend more time adsorbed to or partitioned within the stationary phase and thus take longer to emerge from the column.

- Main components of gas chromatography instrument
  - Carrier gas supply
  - Injection port
  - Column
  - Column oven
  - Detector
  - Recorder
- The analyte is loaded into the head of the GC column via a heated injection port where it evaporates
- It then condenses at the head of the column
- The oven temperature is then either held constant or programmed to rise gradually.
- A retention gap is placed between the injector and the front of the column to retain contaminants and prevent them from reaching the column.

- Elution is brought about by the flow of an inert gas mobile phase.
  - Does not interact with molecules of the analyte
  - Its only function is to transport the analyte through the column
- Separation is based on
    - differences in boiling points of the solutes and
    - the solutes' interaction with the stationary phase
- The degree and speed of separation of components is governed by several factors, including
  - Temperature,
  - Gas flow rate and
  - The physicochemical properties of the individual components as well as those of the stationary




Carrier gas in
Silicone rubber septum
Injector port
Injector oven
Exit
Computer
Detector
Detector oven
Column
Column oven



The chromatogram shows:

1. The **order of elution** (order of components coming off the column) related to boiling points and polarities of the substances in the mixture,
2. The **retention time** (time of elution), and
3. The **relative amount** of the components in the mixture.

Components of GC

- Injection port –
  - Syringe containing sample is introduced through a septum,
  - Injection port oven temperature heated to temperatures that ensures fast volatilization of sample, i.e. above the b.p. of all sample components.
  - Decreased resolution result from slow injection of a large sample volume.
- Carrier Gas –
  - Sweeps analyte to separation column. Usually, He, or N_2, sometimes H_2.

- **Carrier gas**
  - In GC the mobile phase is a gas and is generally referred to as the carrier gas.
  - The three commonly used carrier gases in GC are nitrogen, hydrogen and helium.
  - The choice of the carrier gas is usually dictated by the detector requirements.
  - Helium is the gas of choice for thermal conductivity detector (TCD), providing greater sensitivity than nitrogen.
  - The electron capture detector (ECD), are more efficient when nitrogen or argon-methane mixtures are used as carrier gas.
- **Ideal properties of carrier gas**
  - It must be inert with respect to both sample and stationary phases at the operating temperature.
  - It must be readily available in high purity and at reasonable cost.
  - It must be pure as any impurities may contribute to a background signal at the detector, increasing the noise of the system and there by reducing sensitivity.

- For example:
  - Traces of hydrocarbon can lower detector sensitivity(FID).
  - Trace amounts of water can desorb contaminants in the column leading to high background signals.
  - Traces of oxygen can cause degradation of certain liquid phases such as polyglycerol and polyamides, resulting in changes of solute retention times.
  - Note that moisture and oxygen can be removed with special filters.
- It should not contribute significantly to the detector signal and it should be safe to use.

- Column –
  - Either packed (old) or capillary (newer) columns.
  - Each type requires its own set of plumbing components (threads, etc.)
  - Column T –
    - May be below b.p. of sample components, but high enough to keep a significant quantity in the gas phase.
- Detector –
  - Ideally, it should be sensitive to anything that passes by.

Detectors

- There is not noticeable difference in sensitivity between nitrogen and helium using flame ionization detector (FID).
- Thermionic detector (TD) such as the nitrogen-phosphorous detector (NPD) utilizes nitrogen or helium as the carrier gas.
- Similarly, the photoionization detector (PID) uses oxygen-free nitrogen or helium.
- Nitrogen is used as carrier gas with the flame photometric detector (FID).
- Infrared spectroscopy
- Electrical conductivity

Sample application

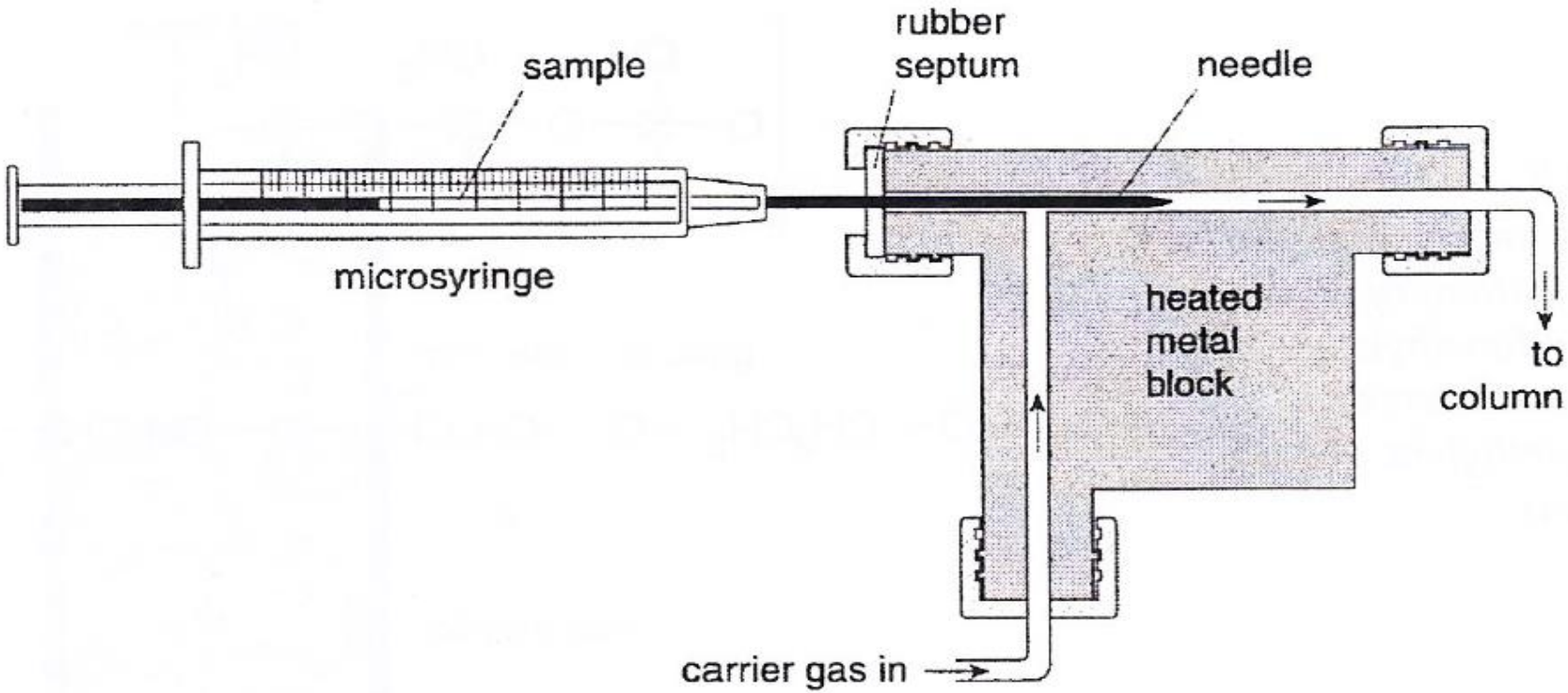
- Samples could be in the form of liquid and gas
- Liquid samples could be applied manually or automatically
- For manual application syringes are used, which can deliver precise amounts in the order of micro liters
- Automated injection systems facilitate batch processing of large numbers of samples
- Automatic samplers considerably enhance injection precision and circumvent the need for flushing the syringe between samples to avoid cross contamination during manual injection
- For gaseous samples, application is done by sampling loops.
- Only very small amount of sample is applied in GC:
  - for liquid samples: 0.1-10 µl
  - for gases: 0.5-5 ml

Injection ports (Sample inlets)

- The function of the sample inlet is
  - to evaporate the sample and
  - transfer it to the column as a compact ”plug”
    - in the shortest possible time and
    - in an accurate and reproducible manner
- The temperature at the injection port is high
- Sample decomposition must not occur during vaporization process
- The injection port must deliver a very small amount of sample to the column to minimize peak width broadening and overlap of peaks.
- This is achieved by splitters

Injection Port

- Sample sizes for standard GC procedures typically involve 0.1 to 10.0 microliters of analyte solution injected into a **heated sample port**.
- The needle must be inserted **carefully** to avoid bending or breaking.



Flow control

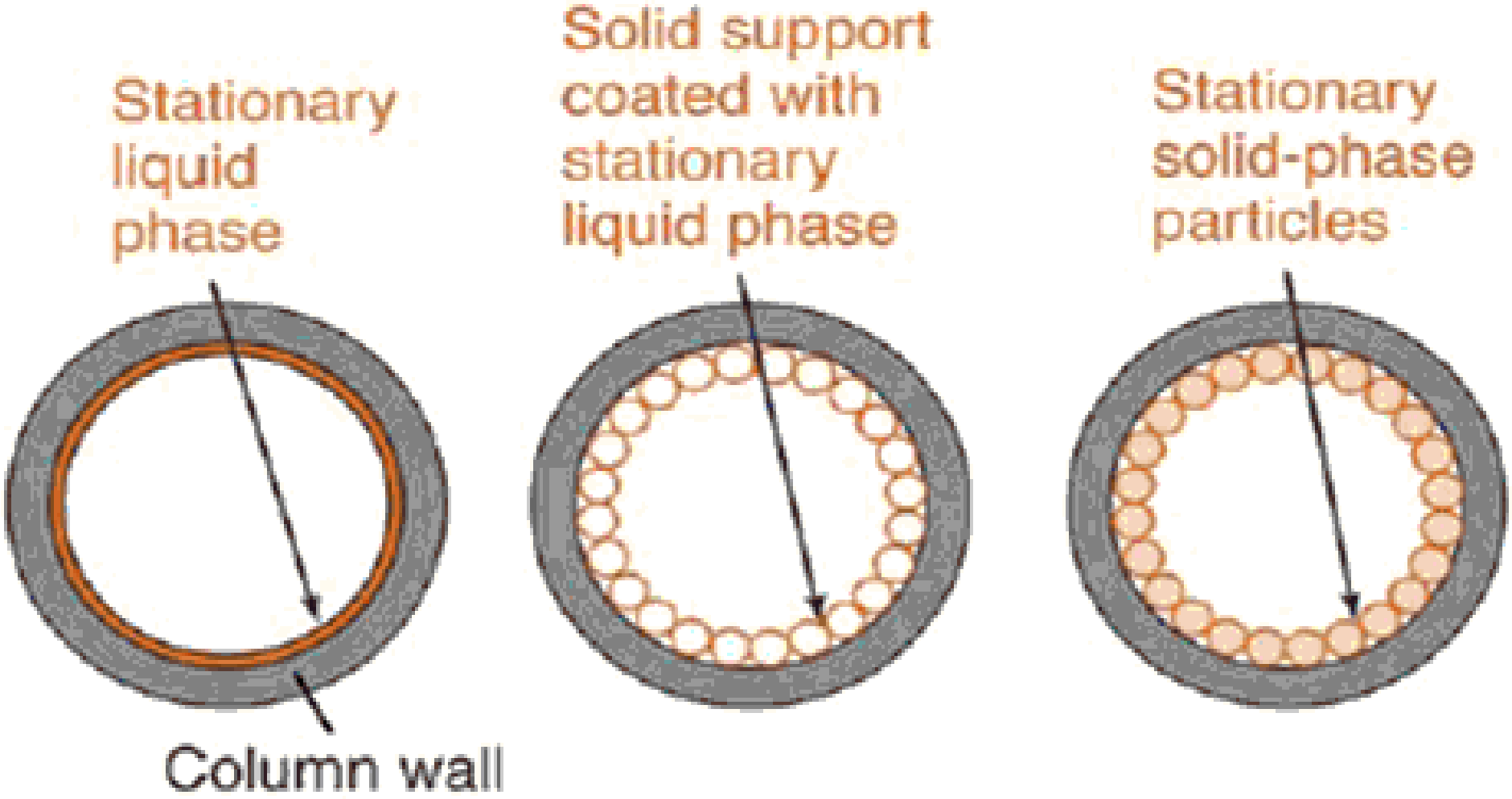
- The carrier gas is feed into the GC via pressure regulator, and flow controllers are used to control the mass-flow rate
- Flow rate is usually made 1-10 ml/min
- accurate and constant carrier–gas flow rate is essential for solute elution reproducibility
- Normally, gas flow rate decrease because of an increase in gas viscosity and column backpressure with an increase in temperature, especially during temperature programmed work
- Differential flow controllers are thus essential to ensure a constant mass flow rate independent of the column resistance
- Detectors usually also require certain range of gas flow rate
- Gas flow rates can be simply measured at the end of the column with soap–bubble flow meter
- Flow is regulated by the action of pressure regulators operating against flow restriction.

Columns

- A chromatographic column physically supports and retains the stationary phase
- Columns are usually made of copper or glass
- The nature of the column influences
  - the amount of sample that can be handled,
  - the efficiency of the separation ,
  - the number of analytes that can be easily separated, and
  - the amount of time required for the separation.
- Two categories of columns are used for gas chromatography.

- Packed columns,
  - which contain particles of the stationary phase packed into a metal or glass tube.
  - A support material is coated with a liquid or semi-solid stationary phase and packed in the column for GLC, or small particles of a solid adsorbent is packed in columns for GSC.
  - The support material could be a bead, porous or porous layer material.
    - A packed column generally has an internal diameter of 2-10 mm and a length of 1-4 meter.
      - decreasing the column internal diameter Increases resolution
    - To fit into the temperature-controlled oven in the gas chromatography, the packed column usually must be carefully bent or coiled.

- Open tubular columns (capillary columns),
  - which do not contain any solid particles to restrict flow (as in a packed column).
  - Instead, the inside of the column wall serves as the support for the liquid phase
  - Because the column is an open tube, there is a very small pressure drop
  - Their internal diameter is 0.1-0.5 mm and may have a length of 10-100 m.
  - There are two main types of open tubular column:
    - The wall–coated open tubular capillary column (WCOT); and
    - the support–coated open tubular column (SCOT)


Stationary liquid phase
Solid support coated with stationary liquid phase
Stationary solid-phase particles
Column wall
Wall-coated open tubular column (WCOT)
Support-coated open tubular column (SCOT)
Porous-layer open tubular column (PLOT)

Effect of temperature in GC

- Injection port temperature:
  - Should be hot enough to vaporize the sample rapidly,
  - Should be low enough to avoid thermal decomposition or rearrangement reaction of the sample
  - Should be at least 50°C more than column temperature

- Column temperature:
  - High enough so that analysis is accomplished in reasonable time and
  - low enough so that separation is achieved
  - the columns must be heated to a temperature at which all the substances are vaporized
  - Column temperature could be
    - constant through out the operation (isothermal analysis) or
    - can be programmed to increase gradually

- The choice of column is dictated by the nature (volatility and polarity of the sample
- Columns have a temperature limit that is normally labeled on their package
- Excessive temperature (above the temperature limit) could result in column bleeding, which is the removal of the stationary phase coat from the support material
- The column is heated to
  - prevent analyte condensation within the column
  - control elution of the different analytes
  - reduce band broadening to get sharper peaks

Samples for GC

- The samples most often analyzed by gas chromatography are
  - those which are liquids at room temperature and which vaporize with out degradation
- Such samples are usually injected with a 1.5 or 10µl-graduated syringe through a septum into a heated, flash evaporation injection system
- The sample is rapidly vaporized, since the injector temperature is held above the temperature of the highest boiling components in the sample
- The carrier gas flushes the vaporized sample from the injector into the column.

Detectors

- After the components of a mixture are separated in the chromatographic column, they must be detected at the outlet so that they can be identified and measured
- Detectors used in conjunction with gas chromatography include
  - Flame ionization (FID),
  - Electron capture detector (ECD),
  - Thermal conductivity detector (TCD),
  - Mass spectrometric detector (MS),
  - Fourier transform infrared detector (FT-IR) and
  - Atomic emission detector (AES)
- The three most often used detectors are TCD, FID and ECD.

- **Properties of an Ideal Detector**
  - Adequate sensitivity
  - Good stability and reproducibility and easy to use
  - A linear dynamic range that extends over several orders of magnitude
  - A temperature range from room temperature to at least 400°C
  - A short response time that is independent of flow rate
  - A predictable response toward all solutes
  - Nondestructive to the sample

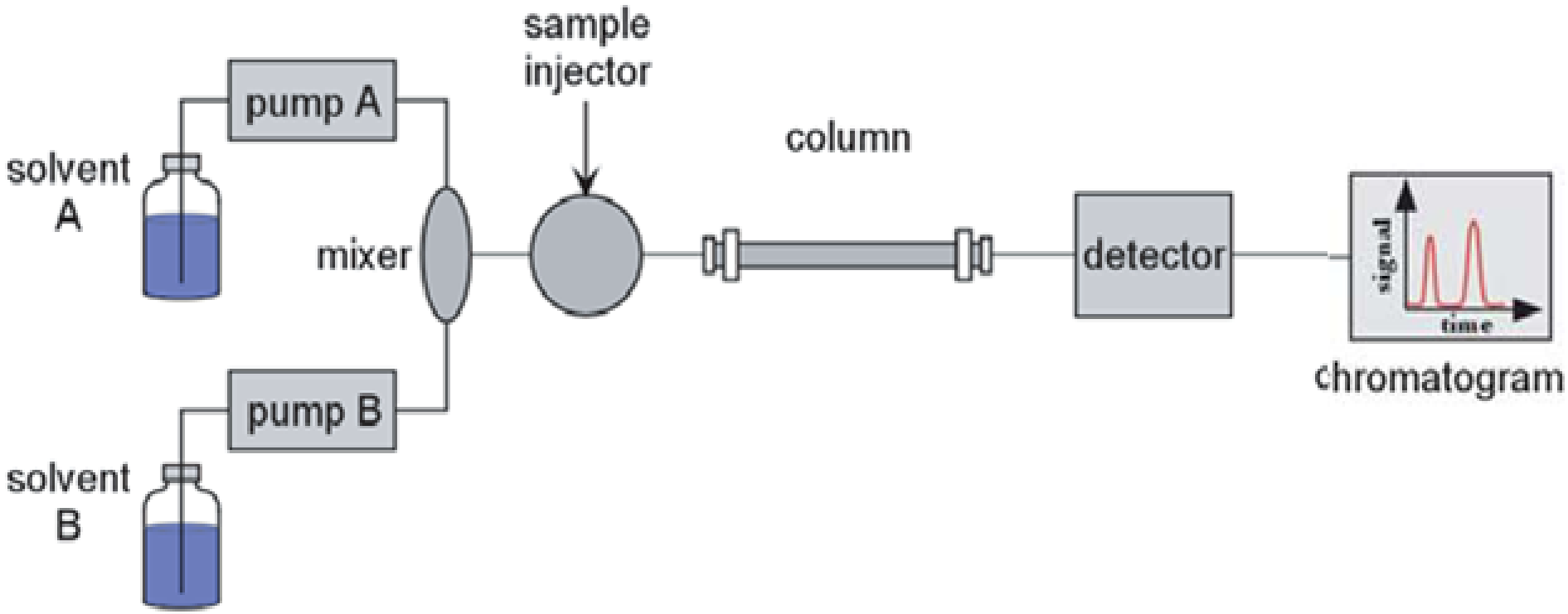
High Performance Liquid Chromatography (HPLC)

- Limitation of simple column chromatography
  - The particle size of the stationary phase can't be less than 150µm
  - Very fine particles will pack tightly, impeded the flow of the mobile phase
    - Acceptable flow rate will not be achieved
- Solution : HPLC
- HPLC is a column chromatography where the mobile phase flows under extremely high pressure, provided by special pumps

- The principal advantages of HPLC compared to classical column chromatography include:
  - **Improved resolution** of the separated substances
  - **Faster separation time**
  - **Increased accuracy**, **precision** and **sensitivity** with which the separated substances may be quantified
  - HPLC allows stationary columns to be reused a number of times without requiring that they be regenerated, and
  - the results of HPLC analysis are more highly reproducible and
  - HPLC permits both instrumentation and quantitation to be automated

- HPLC is a common method used for analysis of
  - Biological compounds
  - Pharmaceuticals
  - Low- or Non-volatile environmental cpds. e.g. DDT
- The main HPLC components are:
  - Liquid mobile phase/solvent reservior
  - Tubing
  - Pump
  - Guard column
  - injection valve (for sampling)
  - column
  - detector and recorder

HPLC Instrumentation



Detectors

- UV
  - Single wavelength
  - Variable wavelength
  - Multiple wavelengths
- Fluorescence
- Electrochemical
- Mass Spectrometric

General outline of the process

- Samples in solutions are separated in a column by a liquid mobile phase(solvent) which is pumped at a high pressure
- The mobile phase is pumped at a constant rate by special HPLC pumps
  - One distinction from other similar techniques
- in the column separation takes place and components are the detected in appropriate detectors
- Chromatogram will be presented as measure of signal intensity versus time

Mobile Phase for HPLC

- Refers to the solvent being continuously applied to the column, or stationary phase
- Acts as a carrier for the sample solution
- Can be altered in order to manipulate the interactions of the sample and the stationary phase.
- Thus, there are several types of mobile phases, which include:
  - Isocratic
  - Gradient

- In **isocratic elution,**
  - Compounds are eluted using constant mobile phase composition.
- In **gradient elution,**
  - Compounds are eluted using mobile phase of variable composition
  - The variation in composition can be made in
    - **Stepwise or**
    - **Linear** fashion.
- The choice of the mobile phase is governed by the physical properties of the solvent
- The eluent strength is a measure of solvent adsorption energy

- the choice of mobile phase should consider the type of stationary phase employed
  - The main distinction, in this regard, is between **reverse** phase and **normal** phase chromatography
  - In **normal phase system**, non-polar solvents such as hexane or iso-octane are used where as **reversed phase** requires polar solvents such as water, acetonitrile or methanol
- Factors to consider include:
  - polarity,
  - miscibility with other solvents,
  - chemical inertness
- The mobile phase must be free from dust
  - This is achieved by a filtering element to the end of the tube leading from the reservoir to the pump
  - This removes particles that Interfere with detection

- Use only HPLC grade solvent; impurities give rise to noisy baselines
- Dissolved air(Oxygen, Nitrogen) should be removed
  - because gases form bubbles and this can cause irregular pumping action and fluctuation signals from the detector
    - Interfere with detection and Cause band broadening
- The dissolved air is removed using Degassers
  - Vacuum pumping of the mobile phase
  - Sparging of the mobile phase with inert gas like helium
  - Agitating the mobile phase in an ultrasonic bath
- When using UV detectors,
  - consideration of the UV absorption of the mobile phase is needed
  - This is indicated by the UV cut-off value

HPLC Pumps

- The main purpose is to pass a constant flow of mobile phase through the HPLC column at pressure ranging 1-550 bar
- It must be special pump that is capable of generating a very high pressure
- Considerations concerning pump are:
  - component of the pump must be resistant to corrosive chemicals and solvents
  - The pump should provide high pressure
  - Flow should be pulse free and stable
- There are several types of pumps available for use and include:
  - Reciprocating Piston Pumps,
  - Syringe Type Pumps, and
  - Constant Pressure Pumps.

HPLC Injection system (unit)

- Two basic types
  - Those in which the sample is injected directly in to the column
  - Those in which the sample is deposited before the column inlet & then swept by a valving action in to the column by the mobile phase

Column

- Chromatographic separation in HPLC system occurs in the column
- Injection ⇒ (pre-column) ⇒ column ⇒ (post-column) ⇒ elution ⇒ detection

A. Separation Column

B. Other columns

- They are:
  - » Guard,
  - » Derivatizing,
  - » Capillary,
  - » Fast, and
  - » Preparatory Columns.

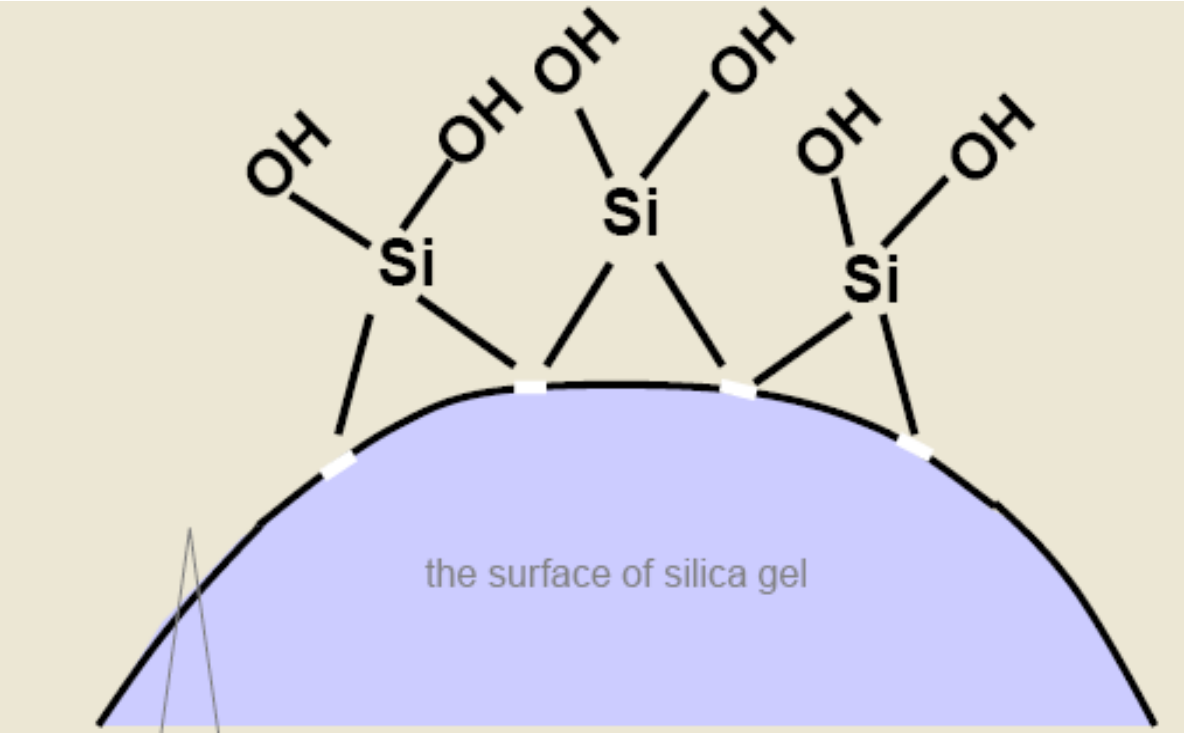
Guard Columns (Pre-column filters)

- Are placed anterior to the separating column.
- This serves as a protective factor that prolongs the life and usefulness of the separation column.
- They are dependable columns designed to filter or remove:
  - particles that clog the separation column;
  - compounds and ions that could ultimately cause
    - "baseline drift",
    - decreased resolution,
    - decreased sensitivity, and
    - create false peaks;

HPLC - Column

✓Packing material

- The most popular is silica gel.
- It is believed that silanol radicals (-Si-OH) on the surface of silica gel act as the active site of sample in separation



HPLC - Column

B. Other columns

- There are various columns that are secondary to the separating column
- They are:
  - ✓Guard,
  - ✓Derivatizing,
  - ✓Capillary,
  - ✓Fast, and
  - ✓Preparatory Columns.

Elution and detection

Elution from the Column

- After the sample is flushed or displaced from the stationary phase, the different components will elute from the column at different times.
  - The components with the least affinity for the stationary phase (the most weakly sorbed) will elute first, while
  - Those with the greatest affinity for the stationary phase (the most strongly sorbed) will elute last.

Detection

- The different components are collected as they emerge from the column.
- A detector analyzes the emerging stream by measuring a property which is related to concentration and characteristic of chemical composition.
- For example, the refractive index or ultra-violet absorbance is measured.
- Two general types of HPLC detectors are used.
  - » **Bulk property or general detectors** measure a change in some over all physical property of the mobile phase plus the solute (e.g. refractive index). These detectors are generally nonspecific and respond to a wide range of compounds;
  - » **Solute property or selective detectors** are sensitive only to some property of the solute (e.g. UV absorption).

- An Ideal detector
  - Should have good sensitivity to all eluting components.
  - Should be reasonably linear so that it can be used in quantitative analysis.
  - Should not significantly degrade the separation obtained in the column.
  - Should be reliable and easy to operate.

- **Recorders and data systems**
  - Recorders and data systems convert the electronic signal out put from HPLC detector into a permanent record for future use.
  - These could be a charte recoreder where a chromatogram plotted on a piece of chart paper or a personal computer.

Summery and Application of HPLC

HPLC pump continually pumping mobile phase from solvent reservoir

⇓

Sample dissolved in mobile phase is introduced into the flowing mobile phase via injector

⇓

Separation of components in column

⇓

Separated components detected by detector

(After detection the eluate is wasted or sent to a fractions collector)

⇓

Detector transmits signals to recording device to give the chromatogram

⇓

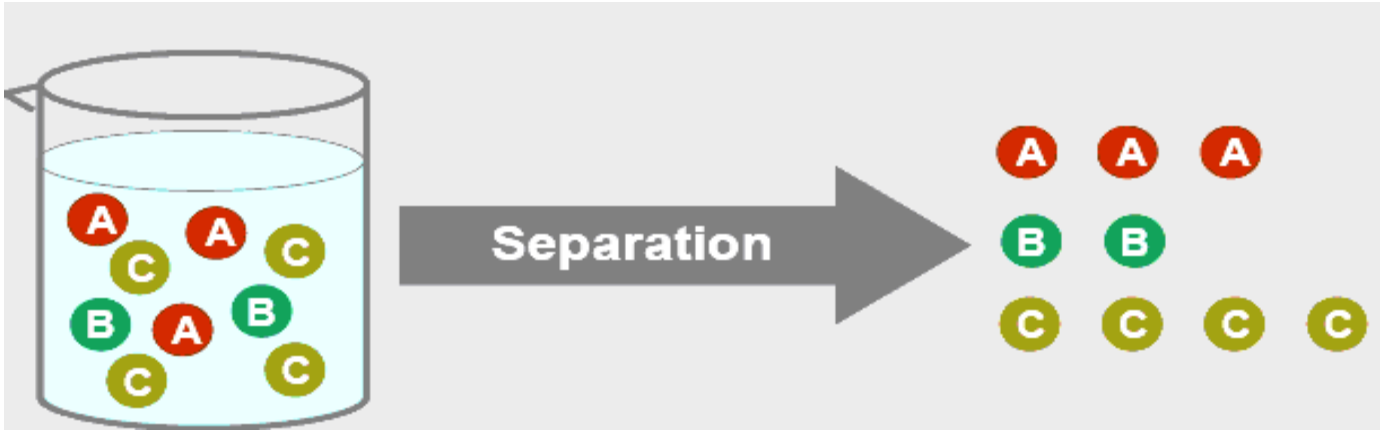
Analysis are done (qualitative, quantitative etc)

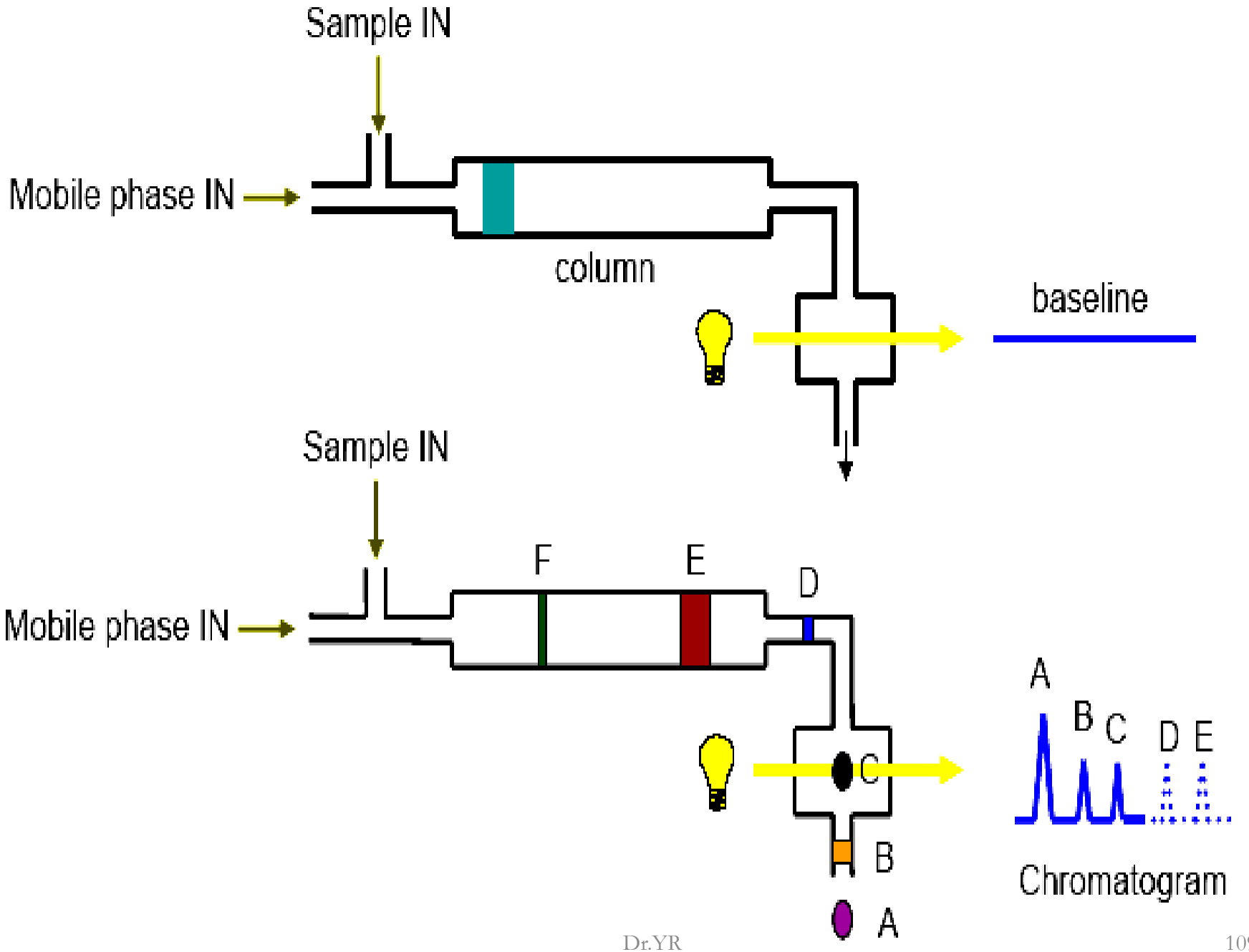
What is HPLC used for?

- ❖Separation of mixed components
- ❖Qualitative analysis / Quantitative analysis
- ❖Preparation of interest components

- Separation can be followed by
  - analysis and/or
  - preparation of interest components

➢**Separation and Analysis**



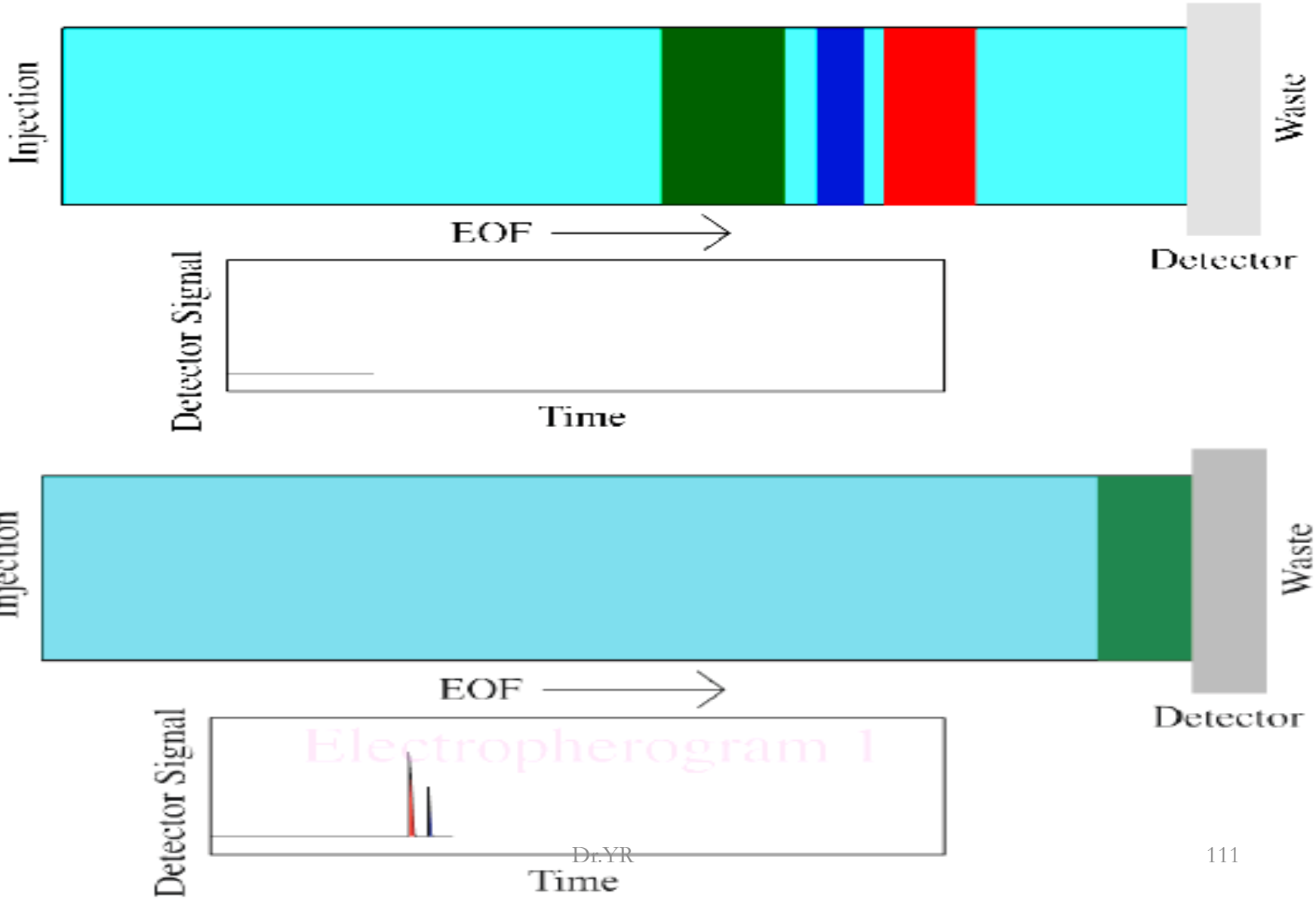

Sample IN
Mobile phase IN
column
baseline
Sample IN
F
E
D
Mobile phase IN
C
B
A
A
B C
D E
Chromatogram

Zone electrophoresis

- **Capillary electrophoresis** (**CE**), also known as capillary zone electrophoresis (CZE), can be used to separate ionic species by their charge and frictional forces and mass.
- In traditional electrophoresis, electrically charged analytes move in a conductive liquid medium under the influence of an electric field.
- Introduced in the 1960s, the technique of capillary electrophoresis (CE) was designed to separate species based on their size to charge ratio in the interior of a small capillary filled with an electrolyte.

Capillary Zone Electrophoresis (CZE)

This represents the entire CE capillary. The potential voltage is applied across this capillary.

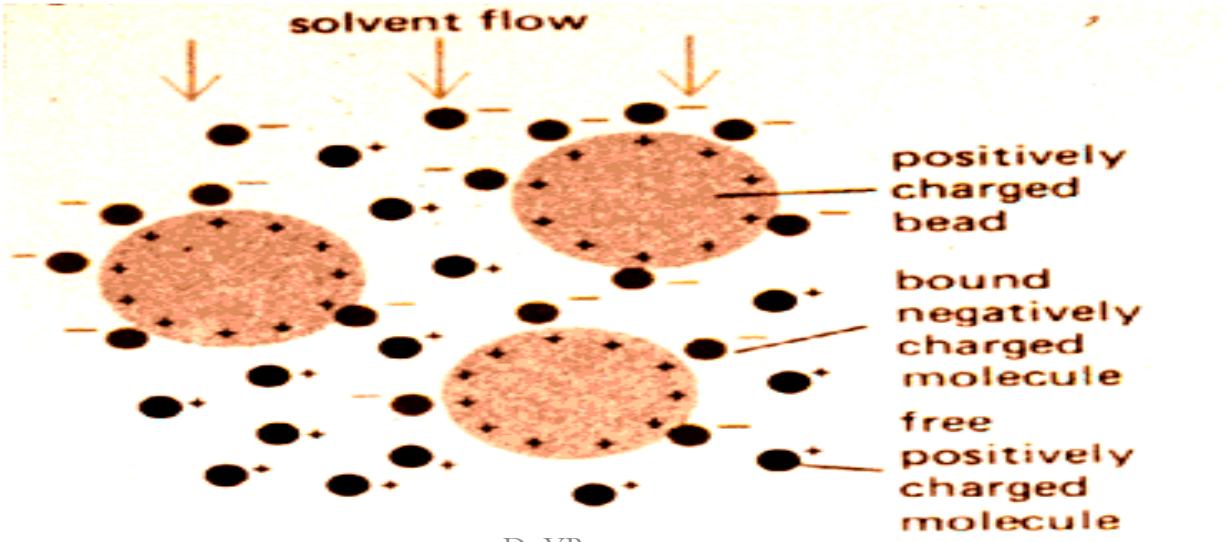


Electropherogram 1

Ion exchange chromatography

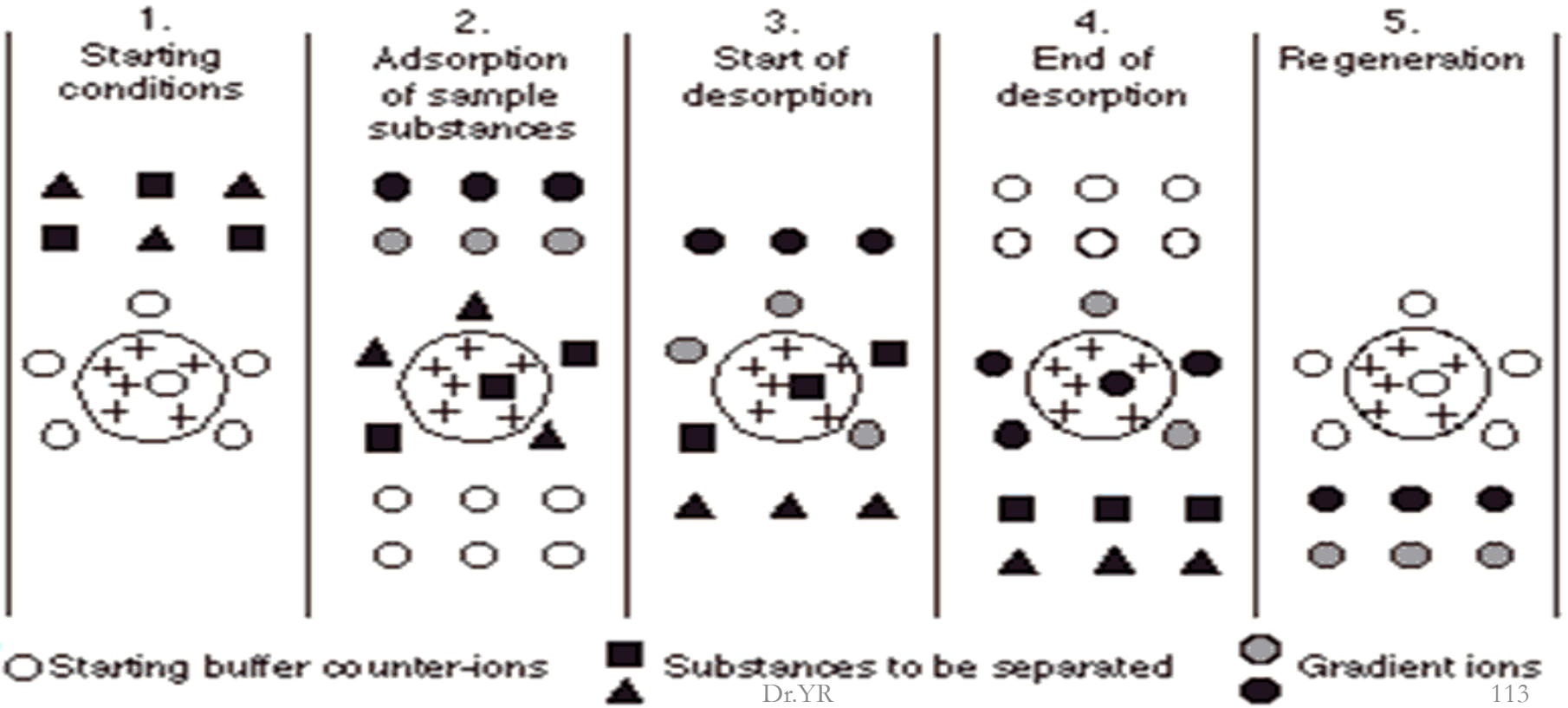
Theory

- Separation in ion exchange chromatography depends upon the reversible adsorption of charged solute molecules to immobilized ion exchange groups of opposite charge.
- Most ion exchange experiments are performed in five main stages.
- These steps are illustrated schematically below.

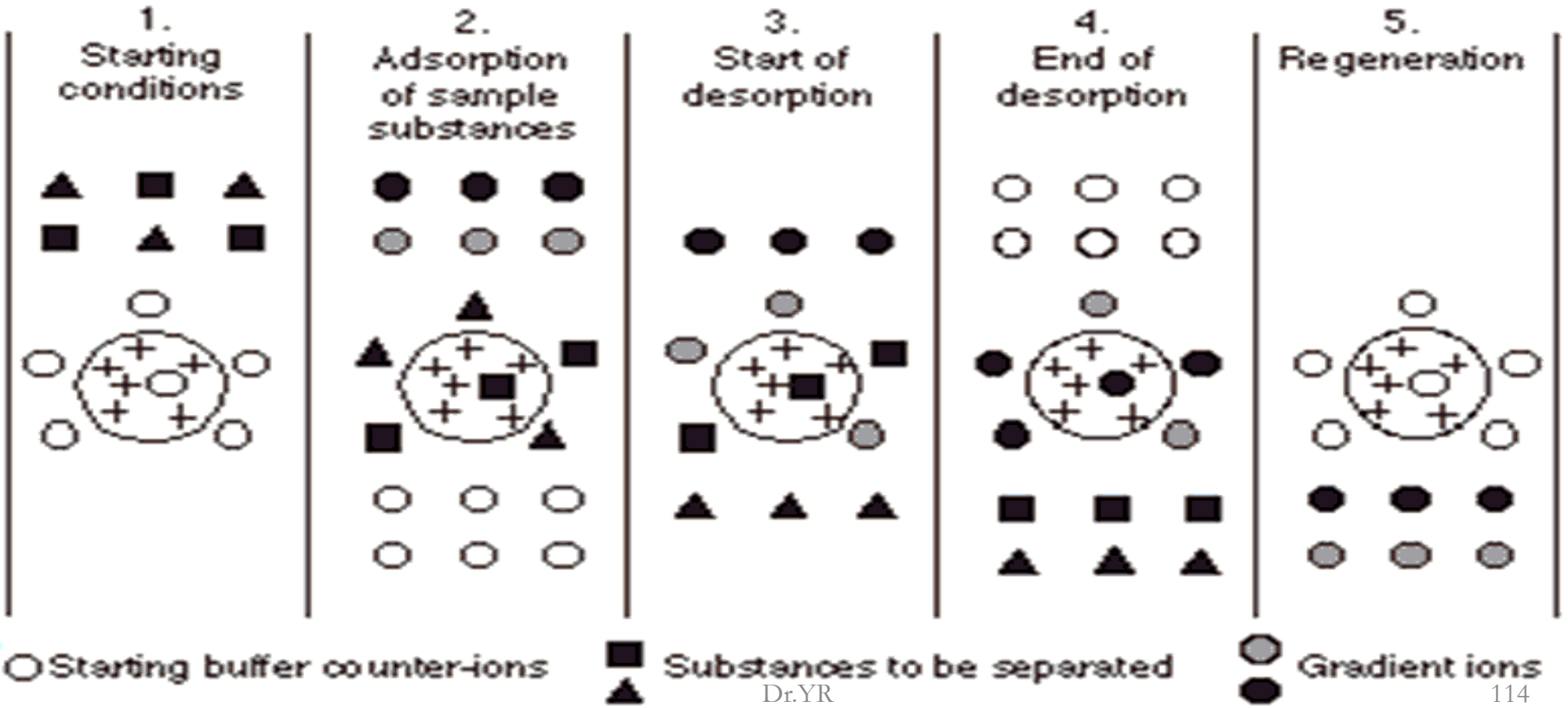


(A) ION-EXCHANGE CHROMATOGRAPHY

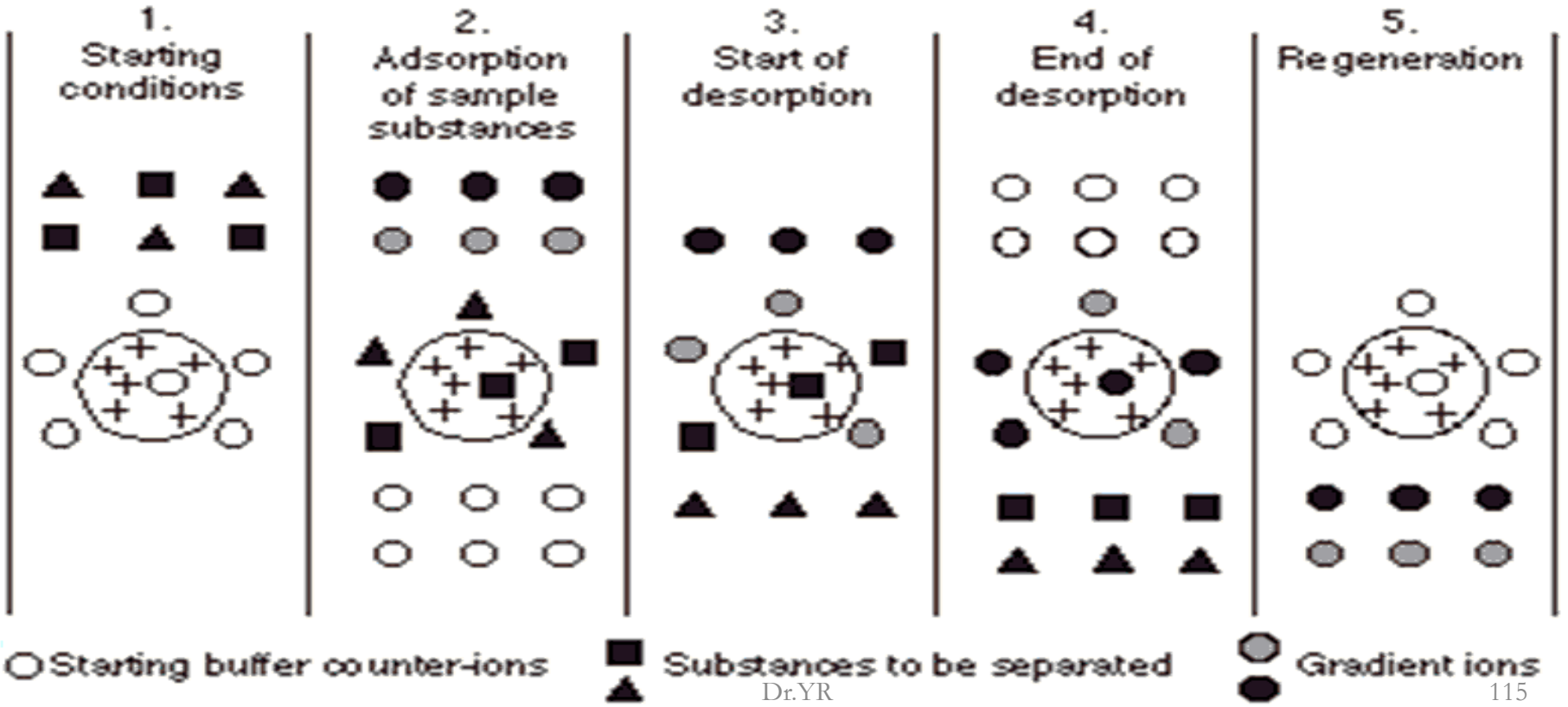
- The first stage is equilibration in which the ion exchanger is brought to a starting state, in terms of pH and ionic strength, which allows the binding of the desired solute molecules.
- The exchanger groups are associated at this time with exchangeable counter-ions (usually simple anions or cations, such as chloride or sodium).



- The second stage is sample application and adsorption, in which solute molecules carrying the appropriate charge displace counter-ions and bind reversibly to the gel.
- Unbound substances can be washed out from the exchanger bed using starting buffer.



- In the third stage, substances are removed from the column by changing to elution conditions unfavourable for ionic bonding of the solute molecules.
- This normally involves increasing the ionic strength of the eluting buffer or changing its pH.
- In Figure 1 desorption is achieved by the introduction of an increasing salt concentration gradient and solute molecules are released from the column in the order of their strengths of binding, the most weakly bound substances being eluted first.



- The fourth and fifth stages are the removal from the column of substances not eluted under the previous experimental conditions and re-equilibration at the starting conditions for the next purification.

